Isabella Handbury

Assistance Chair

Unit 3,4 Product Design

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End User Profile

Name: Melissa Handbury

Gender: Female

Age: 51

Location: Healesville, Victoria

Occupation: Charted Accountant

Background information:

The end-user Melissa Handbury, age 51, is an accountant that works from her home in Healesville Victoria. Melissa lives with her husband and three children along with their cat and dog. Her youngest son Logan has Autism level 3 together with an intellectual disability and epilepsy as a secondary condition. Logan is only 13 but is exceptionally tall and will continue to grow. He is quite affectionate and loves cuddles with his family.

Living Residence:

Melissa lives in a modern home on top of a hill in Healesville Victoria, her home has been recently renovated and is now follows a contemporary aesthetic. The house has a large living room where the family spends most of their time together. The living room has an extensive amount of natural light with ceiling to floor windows. Majority of the furniture in the house is modern, with a few antique pieces. To complement the white pallet of the home, bright artwork is hung on some of the walls. The house is two stories with the bedrooms being upstairs. When her children are not in their rooms, they can be found in the living room which is adjacent to the kitchen.

Hobbies and interests:

The end user's hobbies are limited due to their busy schedule; however, they have a variety of different interests. Melissa is an avid collector of Dinosaur Designs which are unique hand-crafted resin jewellery and homewares. Melissa's collection is prominently displayed in her home in a location visible from the living room. When Melissa is not working in the evening, she enjoys sitting in the living room and playing match-three puzzle games while watching Netflix. Melissa also loves plants and takes great pride in the garden surrounding her home. She has a small greenhouse housing her succulents and cacti, she spends her limited free time tending to her plants.

Career:

The end user works from home as a charted accountant, she has extensive experience in both private practice and the commercial sector. Melissa is the founder and Chair of Lyrebird college, an independent co-educational specialist school for children with an Autism Spectrum Disorder.

Extra Information

Autism:

"Logan's diagnosis of ASD (Autism Spectrum Disorder) is a lifelong condition. The severity of Logan's ASD diagnosis is level 3: he requires very substantial support in developing verbal and nonverbal social communication skills, coping with change, emotional regulation and behavioural development. He experiences severe difficulties in verbal and non-verbal communication, very limited speech, as well as odd and repetitive behaviours.

Logan experiences significant difficulties across a number of areas affecting his capacity to function effectively at home and school, interact appropriately with others, manage his emotions and behaviour, and form friendships with peers.

Logan has significantly underdeveloped receptive and expressive language skills. He is unable to follow more than two-step instructions and can be easily distracted by his surroundings. Within busy environments, Logan requires an adult to be within arms reach to increase the likelihood he will process single verbal instructions.

Self-stimulatory behaviours: Logan has a wide range of sensory needs which fluctuate depending on external factors including sleep and mood. Logan is sensitive to the feel of tags, seams in his pants and socks…

Logan also demonstrates sensory seeking behaviours. He frequently seeks deep pressure (E.g. someone laying across him), pushing against heavy items or a wall, water, and motion. "

-Logan's most recent phycology report.

Epilepsy:

Description: "Tonic-clonic seizure – if standing Logan will fall down. His body will stiffen and begin to convulse. During his seizure he will make a weird gurgling noise. His eyes will be wide open and may roll back. There is also excess saliva causing him to droll."

Typical duration: 2-3 minutes.

Known triggers: none

Support during a seizure: Where possible place him in the recovery position and place something under his head for support. Ensure that Logan is safe, and his airway is unobstructed.

-Logan's epilepsy management plan

What is Autism?

About Autism Spectrum Disorder (ASD)

- Children with Autism Spectrum Disorder (ASD) exhibit a range of impaired social, communication and cognitive skills that can significantly impede their participation in school, their families and society overall.
- ASD affects 1 in 100 Australian children – this also means ASD impacts 1 in 100 Australian families.
- We estimate that as of June 2019, there are over 13,000 Victorian children under the age of 13 who have ASD. This is a conservative estimate based on 2019 population and education statistics.
- ASD prevalence figures are growing. More children are diagnosed with ASD than with AIDS, Diabetes and Cancer combined.
- Boys are diagnosed with ASD 4x times more often than girls. Girls are generally diagnosed with ASD at a later age than boys. They mask their challenges better and can be misdiagnosed with a mental health condition rather than ASD.
- Whilst there is no 'cure' for autism, many children with ASD are capable of significant development with suitable therapeutic support.
- Autism is a wide range spectrum disorder: it is different from individual to individual. Some children with ASD can be very social and achieve academically.

https://www.learningforlife.com.au/about-autism/

Design Brief

The end-user, Melissa has a son Logan who has Autism level 3 together with an intellectual disability and epilepsy as a secondary condition. Logan is only 13 but is exceptionally tall and will continue to grow. He is quite affectionate and loves cuddles with his family. Melissa requires a chair that will take her son Logan's weight and size off her but still ensure that he can have the physical contact. The chair will need to use long lasting and durable materials as Logan is expected to continue to grow and Melissa would like this chair to last a long period of time. The chair needs to be able to withstand repeated everyday use as Melissa intends to use this chair with Logan every day when he gets home from school and is dysregulated and needs physical contact.

The product will be used in the end user's living room where there is the most amount of free space. The end user's living room is exposed to extensive amounts of natural light during the day which is the primary time the product will be used. As such the final product could fit in with the already existing furniture, including light coloured timber paired with black accents and neutral fabrics. Melissa enjoys spending her rare free evenings in the living room and playing match-three puzzle games while watching Netflix, due to this the end user requires the product to be multifunctional and be comfortable for her to use without Logan also using the chair.

The product can possibly have a smooth finish as Logan has a preference to that particular sensory feeling. The product needs to be easy to access as Logan will be getting in and out of the chair consistently. The product must safely withstand Logan rocking and moving without the product collapsing, this is vital for the safety of both end users.

The budget for this product is $1500, but the end user is willing to move on this to achieve a clever design and quality product that meets the requirements and will be long lasting. Melissa would like the product finished by November in time for school holidays where the product will get more frequent use but is flexible on the date.

Purpose, Function, Context

User-Centred Design

Visual, Tactile, Aesthetic

Innovation, Creativity

Materials

Technologies

Economics

Sustainability

Legal Responsibilities

Constraints

- Must be long lasting and durable
- Must support both end users' combined weight
- Must account for Logan's expected future growth
- Must be durable for repeated everyday use
- Must be multifunctional to use with and without Logan
- Needs to be easy to access
- Must be safe for both end users' weight

Considerations

- Could aesthetically fit in with preexisting furniture
- Could have a smooth finish
- The budget could be $1500 and is flexible
- Could be finished by November

Quality Statement

The end user, Melissa, expects the final product to be finished to a high standard. The finished product should be carefully constructed with durable and long-lasting quality materials to ensure the product can support the required weight. Any wooden materials that are used in the product should be finely sanded and glossed for a smooth high-quality finish.

Evaluation Criteria

Criteria	Why is this important?	How to achieve this is the design and production stages?	How will this be tested with the final project?	Importance (1-10)
Constraints				
Is the final product long lasting and durable?	This is important as the end user has requested that the product be durable and long lasting by implementing suitable materials. This is needed as Logan is expected to continue to grow and the product needs to accommodate for that.	This can be achieved during the design stage by researching a variety of different materials and testing their suitability. During the production stage multiple tests will be done to ensure the durability of the product including running tests with the end users.	The final product will be long lasting and durable, this will be tested by checking in with the end user after delivering the product to determine if there are any concerns about the product's durability.	9/10
Does the final product support both end users' weight?	This is important as the purpose of the product is to accommodate both end users to take Logan's weight off Melissa whilst still allowing for physical contact. The final product needs to be able to allow for Melissa and Logan to both spend time together in the chair.	This can be achieved during the design stage by gathering the end users' anthropometric data and designing the product to these specifications. During the production stage this can be achieved by doing many tests on the product to ensure it can handle the required weight.	The final product will support both end users' weight, this will be ensured after numerous tests during the production stage and a final test with the end users once the final product is delivered.	10/10
Is the final product durable for repeated everyday use?	This is important as Melissa intends to use this chair with Logan every day when he gets home from school and is dysregulated and needs physical contact.	This can be achieved during the design stage by researching long lasting materials and building techniques. These can then be implemented into the design. During the production stage the product will be constructed with repeated everyday use in mind. Ensure that all joins and finishes are done to increase the durability of the product.	The final product will be durable for repeated everyday use. This will be tested by monitoring the condition of the final product after the end users have used the product.	9/10
Does the product account for Logan's expected future growth?	This is important as Logan is only 13 but is exceptionally tall and is expected to continue to grow. Melissa wants the product to last a long time and to achieve this the product must accommodate Logan's future height and weight.	This can be achieved during the design stage by gathering Logan's anthropometric data and researching into his predicted future growth. The product will be designed to account for his growth and for any unplanned excess growth. During the production stage many check ins will be done with the end users to ensure they are happy with the sizing of the product and making sure the final product has excess amount of room.	The final product will account for Logan's future growth, this will be tested by monitoring Logan's growth over the years with check ins with the end users and relating the measurements back to the ergonomics of the chair. Logan will be able to comfortably use this chair into adulthood.	9/10

Is the final product easily accessible?	This is important as Logan will be getting in and out of the chair consistently and requires ease of access.	This can be achieved during the design stage by researching accessible furniture and designing a product with this function in mind. During the production stage excess unnecessary materials or aspects will be removed if they effect the accessibility of the product, after conversations with the end user.	The final product will be easily accessible, this will be tested by recording the time taken to get in and out of the chair along with user feedback.	8/10
Is the final product multifunctional and can be used without Logan?	This is important as Melissa enjoys spending her rare free evenings in the living room and playing match-three puzzle games while watching Netflix, due to this the end user requires the product to be multifunctional and be comfortable for her to use without Logan also using the chair.	This can be achieved during the design stage by researching and incorporating multi use furniture aspects into the design. Whilst designing, prototyping can be done to ensure the final product will have room to be multifunctional. During the production stage the construction of the product will account for making it multifunctional and discussions will be had with the end users to see that the final product will fit this secondary function.	The final product will be multifunctional and will be able to be used without Logan in the chair. This will be tested after the final product is delivered by having the end user utilise the multifunctional nature of the final product.	7/10
Is the final product safe?	This is important as the configuration of the seating arrangement of the chair will have both end users' weight being supported by the chair. The chair must be safe as to not put the end users in danger.	This will be achieved during the design stage by researching the weight limits, stress and strain breaking points of certain materials. A risk assessment will be done to weigh the major safety concerns of the product. During the production stage the product will go though many simulated tests before being safely tested by the end users.	The final product will be safe for both end users and the end users will feel safe whilst using the chair.	10/10
Considerations				
Does the final product aesthetically fit in with the pre-existing furniture?	This is important as the end user has mentioned that their living room follows a set colour and material theme and it would be nice if the final product fit with this theme.	This will be achieved during the design stage by suppling the end user many designs and asking for feedback on the aesthetic of the designs. During the production stage feedback will be requested from the end user to ensure the product is fitting with their pre-existing furniture.	The product will fit in aesthetically with the pre-existing furniture, this will be determined by conversations with the end users and by viewing the final product in its end location.	5/10

Research Table

Research Area	Why is important?	Summary of findings	References
Materials	The research of materials is important due to them being a main aspect for the durability and aesthetics of the final product. By researching a comprehensive number of materials, the design and production stage become easier.	A variety of materials can be used for this project that have a variety of prices, looks and durability. Metals, plastics, fabrics and cushion padding were looked into, and a variety of types were explored.	https://makeitfrommetal.com/different-types-of-metal-facts-and-uses/ https://www.acplasticsinc.com/informationcenter/r/7-different-types-of-plastic-and-how-they-are-used https://homedecorbliss.com/types-of-cushion-stuffing-filling/ https://www.picocleaners.com/blog/the-12-different-types-of-fabric/ https://www.alumitec.com.au/aluminium-angles.php https://www.metalsupermarkets.com/metals/brass/ https://www.azom.com/article.aspx?ArticleID=16788 https://plasticoceans.org/7-types-of-plastic/ https://www.indiamart.com/proddetail/plain-cotton-fabric-14343415955.html https://polinacouture.com/en/to-understand-all-differences-between-silk-and-satin-and-nomore-be-misled/ https://ozupholsterysupplies.com.au/product/custom-foam-item/
Woods	This is important as different types of woods are more durable and have different densities. Woods also have different colours that appeal to different aesthetics.	There are a multitude of different woods that could be used for the project all having different looks, prices, durability, and textures. Some woods are more aesthetically fitting but have lower durability.	https://www.trada.co.uk/ https://www.woodsolutions.com.au/ https://www.timberblogger.com/ash-wood-properties-and-its-advantages-and-disadvantages/
Joins	This is important as the strength of this chair is vital, and it needs to be durable. The joins will determine how much weight the chair can take and its longevity. By researching the types of joins I am able to attain a better understanding of the strongest and most efficient joinery.	The variety of joins are all good for different things, often the more difficult the join is to complete the stronger it typically is. For this project and its need for durability, lots of time will need to be spent ensuring that the joins are perfected.	https://www.toolstoday.com/global/g-46-dovetail-joints-different-types-and-their-uses https://www.woodsmith.com/article/using-pocket-hole-joinery-jig-types-simple-to-full-featured/ https://www.diydoctor.org.uk/projects/timberjoints1.htm https://kitchencabinetkings.com/glossary/mortise-tenon/ https://www.toolstoday.com/global/g-46-dovetail-joints-different-types-and-their-uses http://www.woodworkbasics.com/biscuit-joint.html https://en.wikipedia.org/wiki/Butt_joint
Steam bending wood	This is something I found interesting and want to investigate further, steam bending the wood will have an interesting aesthetic quality. Along with it being visually appealing is that the curves will be safter and more accessible.	To summarise my findings the main pros to steam bending are that it has minimal waste, is low energy and an economical method of manipulating wood. Some woods are better at bending with steam, and this is related to the grain and density of the wood. Steam Bender Australia supplies a service to steam bend material, this could be a possible interesting aspect of my final product.	https://en.wikipedia.org/wiki/Steam_bending#:~:text=Steam%20bending%20is%20a%20woodworking,will%20hold%20their%20new%20shape https://www.designboom.com/design/bar-gantz-steam-bending-wood-furniture-01-15-2017/ https://en.wikipedia.org/wiki/Steam_bending#/media/File:Ngv,_Gebr%C3%BCder_Thonet,_culla,_1870_circa.JPG https://steambenders.com.au/
Finishes	This is important as the finishes effect the look and feel of the product. A smoother finish has been asked to be considered by	After looking into finishes I have deduced that the type of finish depends on the colour, feel, look and type of wood being used. Some	https://octaneseating.com/blog/wood-finishes/ http://theplywood.com/how-to-finish https://octaneseating.com/blog/wood-finishes/ https://www.woodsolutions.com.au/timber-finishes-interior#:~:text=Commonly%20used%20wood%20finis

	the end user and I wanted to explore this option.	finishes are more ideal of different environments and different finishes need to be applied differently.	hes%20include,amount%20of%20oil%20and%20solvent.
Tools	It is important to do an overview on some possible tools that will be needed for this project, to achieve a better understanding of the methods and functions of relevant tools.	After looking into various tools that will be useful in the production stage of this project, I have a greater knowledge of which tools are best for cutting which materials and the functions of the tools. This research has made it easier to identify each tool in the workshop.	https://www.wikipedia.org/ https://www.wonkeedonkeetools.co.uk/clamps/what-are-the-different-types-of-clamp https://www.machineryhouse.com.au/s816 https://guide.directindustry.com/choosing-the-right-electric-screwdriver/#:~:text=An%20electric%20screwdriver%20is%20a,corresponds%20to%20the%20screw%20head. https://wiki.ideashop.iit.edu/index.php?title=Cordless_Drill#:~:text=A%20cordless%20drill%20is%20an,other%20threaded%20fasteners%20into%20material.
Welding	This is important to research as welding is a possible tool that will be utilised to make the final product as strong as possible. If the final product includes the use of metal, then welding will most likely need to be used.	The main important points I discovered in researching welding is the different types of welding and the best metals to weld with along with the general process of welding.	https://thewerkscandc.com/mig-tig-arc-determine-kind-welding-process-use-job/#:~:text=Three%20of%20the%20most%20common,know%20about%20each%20of%20them.&text=Arc%20welding%20is%20the%20oldest%20of%20these%20three%20welding%20processes. https://en.wikipedia.org/wiki/Welding https://www.safetyhub.com/au/safety-training/oxy-acetylene-cutting-safety/ https://www.millerwelds.com/resources/article-library/gmaw-mig-aluminum-welding-hints https://www.weldclass.com.au/blog/39-the-tig-welding-process-your-questions-answered
Age group growth statistics	This is important as the end user, Logan, is expected to have a large continuous growth and the chair needs to last him into adulthood. By researching this I can factor his expended growth into my designs.	To summarise the research the expected growth of a male in the 90th height percentile is over 190 cm tall.	https://thebonescience.com/blogs/journal/5-height-calculators-for-prediction https://tall.life/height-percentile-calculator-age-country/ https://www.hopkinsmedicine.org/health/wellness-and-prevention/the-growing-child-adolescent-13-to-18-years#:~:text=Adolescence%20is%20a%20time%20for,visible%20at%20the%20same%20time.
Anthropometrics	This is important as the end user's anthropometric data will help to make a more comfortable end product and will need to be factored into the durability and longevity of the product.	Logan's anthropometric data was recorded. Melissa has requested that average ergonomic data be used for her information.	https://www.osha.gov/sites/default/files/2018-12/fy15_sh-27643-sh5_ErgonomicsWorkbook.pdf

Woods

Wood	Durability	Density average Kg/m³	Texture	Price	Typical uses	Appearance
Softwood						
Cedar	High	580	Straight grained, medium to fine	Low	Joinery	
Cypress	High	510	Straight grained, medium	Medium	Flooring, furniture	
Pine (southern)	Medium	670	Medium	Medium	Joinery, structural	
Spruce (Canadian)	Low	400	Medium	low	Structural	
Redwood (European)	Medium	510	Medium	Low	Joinery, furniture, structure, moulding, flooring	
Hardwood						
Ash	Medium	620	Straight grain, medium	Low	Furniture, flooring	
Balsa	Low	160	Fine	Low	Interior joinery	
Birch (European)	Low	670	Fine	Low	Furniture	
Blackwood	High	650	Medium	High	Furniture, steam bending	
Bloodwood	High	1140	Straight grained, coarse	High	Furniture, construction	

Steam bending woods



Steam bending wood is a technique where wood is exposed to steam to make it pliable. The heat and moisture from the steam softens the wood fibres so that they can be stretched and bent. When the wood is cooled down the wood fibre can hold its new shape. Modern steam bending is done is a steam box that makes the wood bend around a form and a metal band is used to support the wood to stop blowouts. This manufacturing process is used to make a variety of products including wooden instruments, boats, and furniture. Steam bending has traditional history dating far back, where the practise was vital for the creating of weapons, tools, and water vessels.



Steam bending is a low energy, ecological and economical method of manipulating wood. This method doesn't require drying time for glues to join woods and is less expensive. Steam bending also leaves less scraps than alternative methods of manipulating wood.

Steam bending has a limit on the degree the wood can bend, especially thick woods. The hardness of the wood determines how long the wood needs to soak in water before it can be bent

Popular woods that react well to steam bending are; hardwoods- maple, cherry, birch, walnut, ash

softwoods – pine, spruce, cedar, redwood

Steam benders Australia

"Steam Benders provide a number of different timber bending options to suit a range of applications, from furniture and lighting, to much larger architectural requirements."

WHY STEAM BEND?

Durability

You can always cut a curved shape out of a square one, but what you may gain in simplicity, you will lose in strength and durability. Both steam bending and lamination have the benefit of manipulating the wood so the grain follows the curve and is continuous across the whole part. This is especially important on structural elements like chair legs and backs, when the curved piece is going to be supporting weight and taking stress....



Economy

Steam bending produces less waste than reductive methods. Bending a single piece rather than cutting the shape from a larger mass, retains the continuous long-grain pattern, there are no glue lines, and it doesn't waste material....

Aesthetic

Steam bending is an impressive technique, offering seamless, durable components. The steam bending process produces finished parts with the best overall appearance, with continuous end-to-end grain and no visible glue line.

15 Steel Rd, Yackandandah Victoria 3749

0412 407 797

info@steambenders.com.au

Growth patterns

The height of someone is determined by a combination of genetics and environmental factors. Studies conclude that genetics is responsible for 60-80% of someone's height. Based on the CDC growth charts that use percentile curves that rank the position of an individual and indicate by referencing past statistics if the individual will be greater or below average for height and weight.

Logan has always been in the 90th percentile which indicated that he will continue to grow to a height of 190-200cm. This data is backed up by Logan's dad being 192cm tall.



Anthropometrics

Logan's data:

Left arm length: 78.5cm

Right arm length: 77.6cm

Torso length: 46.1cm

Weight: 67Kg

Height: 185cm

Shoulder width: 46.2cm

Wingspan: 192.4cm

Knee to back (sitting): 74.2cm

Knee to floor distance: 57.1cm

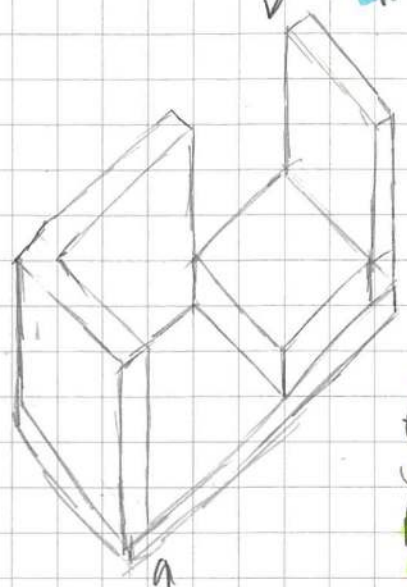
Hip to floor distance: 113.2cm

*The end user (Melissa) has requested that her data be taken from ergonomic averages.

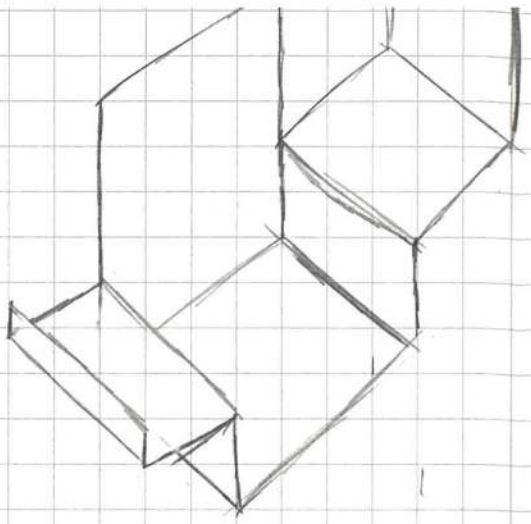
Chair Body

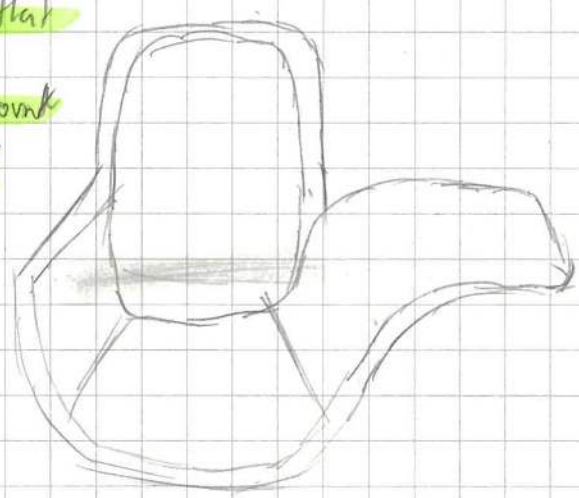
person B

person A



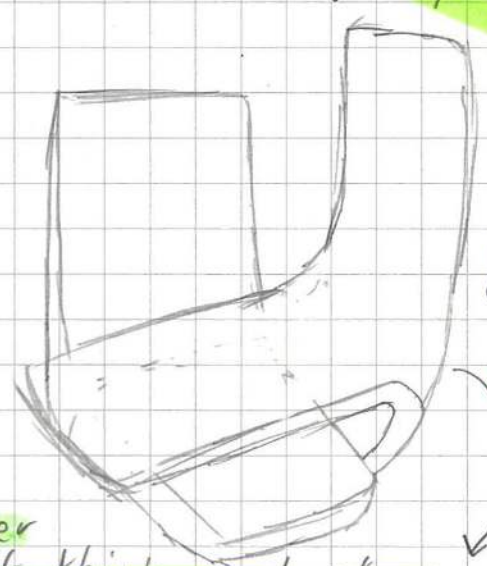
The end user believe that this design would not account for the futur growth of her son.





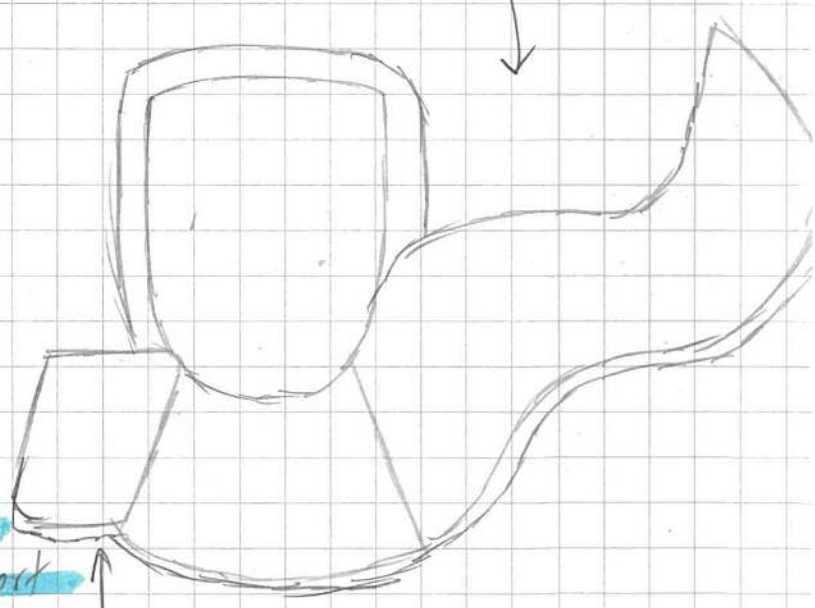
The end user likes the moving element of this design but thinks more back support would be comfortable.

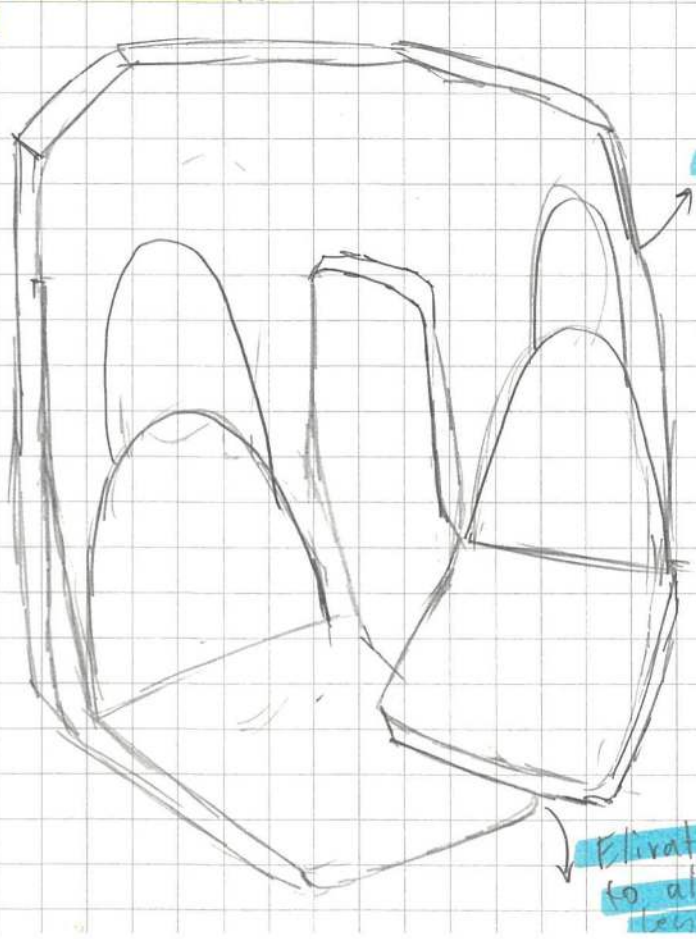
Can rotate down for easy access



End user doesn't think a back on the higher chair is needed.

The end user likes the curve in this design as it is innorative and unique





Back support padding

Foot rest

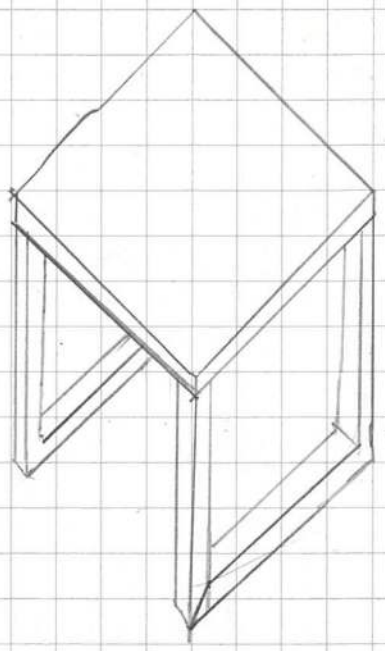
The end user likes the placement of the seating but thinks the design is too bulky.

Elivated to allow for less room

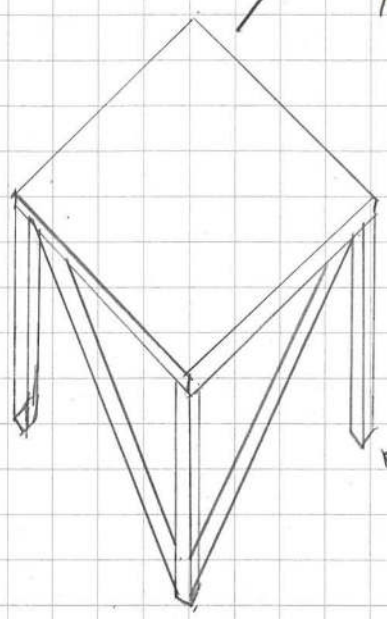
Designer annotations

End user comments

Legs



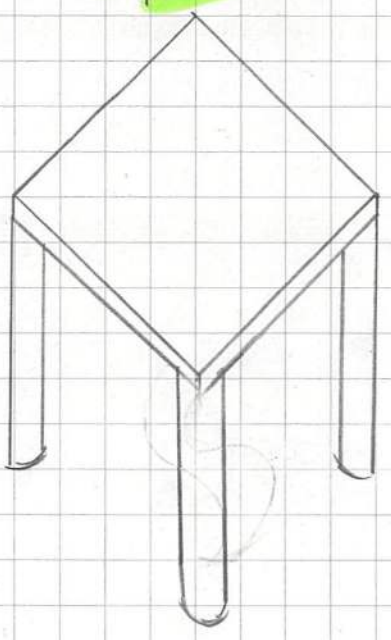
mitre butt join



The lack of symetry would "do my head in" - end user

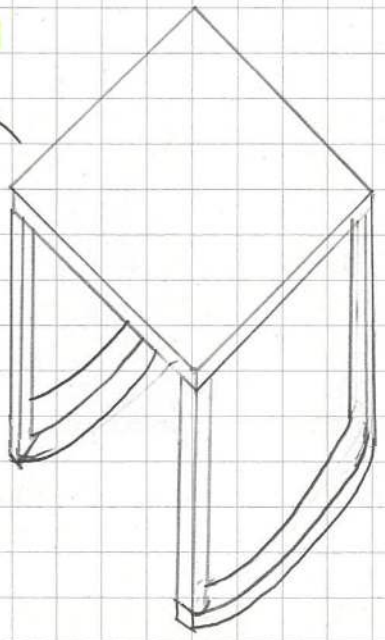
Diagonal support beams for strength

mitre butt join

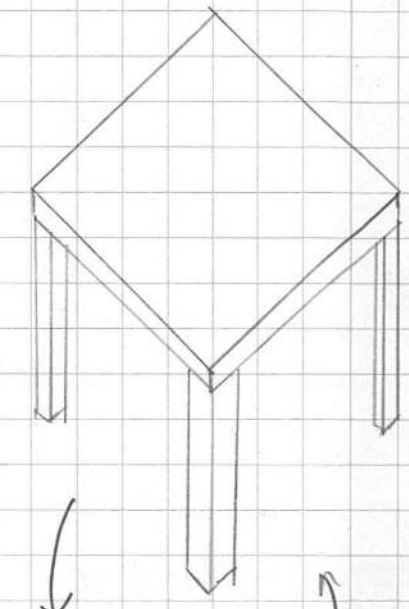


End user thinks that the rocking function isn't as safe for Logan so fears it could damage her floor

curved legs

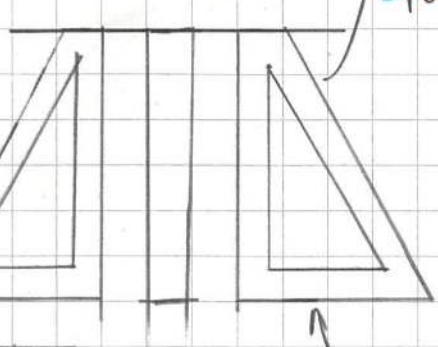


rocking legs to adjust for Logan's rocking.

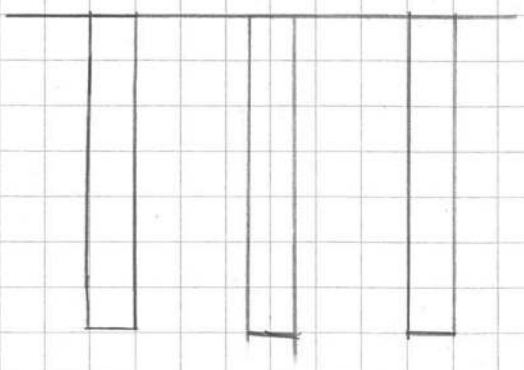


End user thinks these are boring.

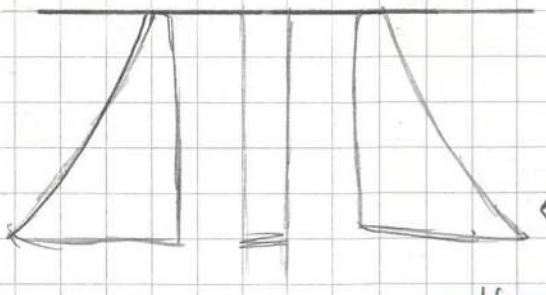
standard chair legs



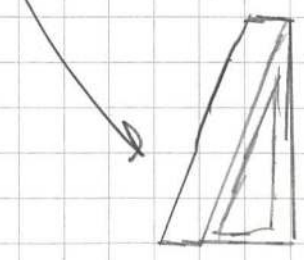
Hollow legs for aethtics



Engraved details?



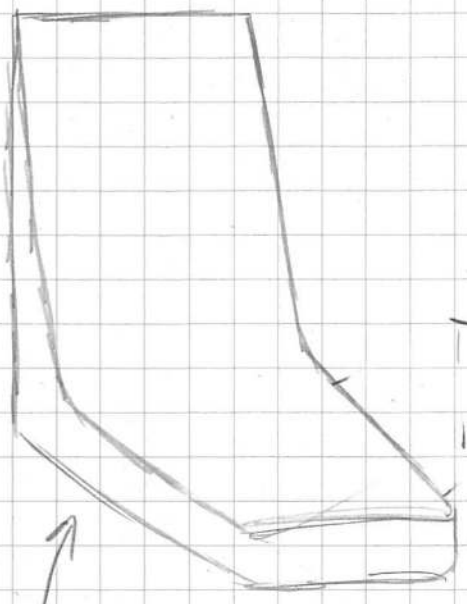
Heavy and excess material



End user likes the aesthetic but needs to see it in more detail.

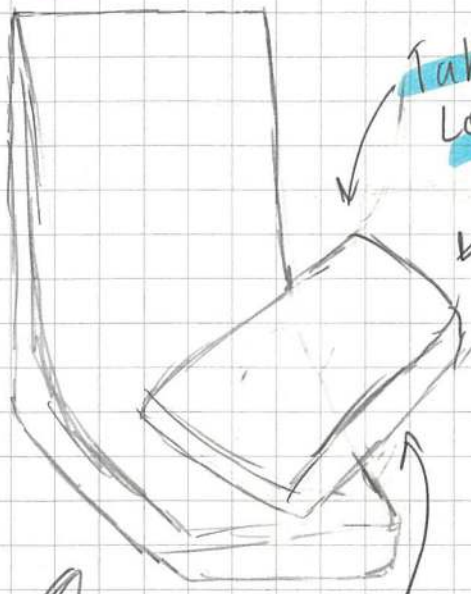
Second Seat

Base



Arm rest

With Assistance Feature



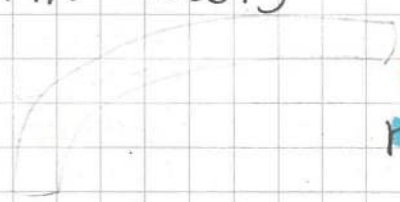
Takes Logan's weight

no backing so Logan can lean on Melissa

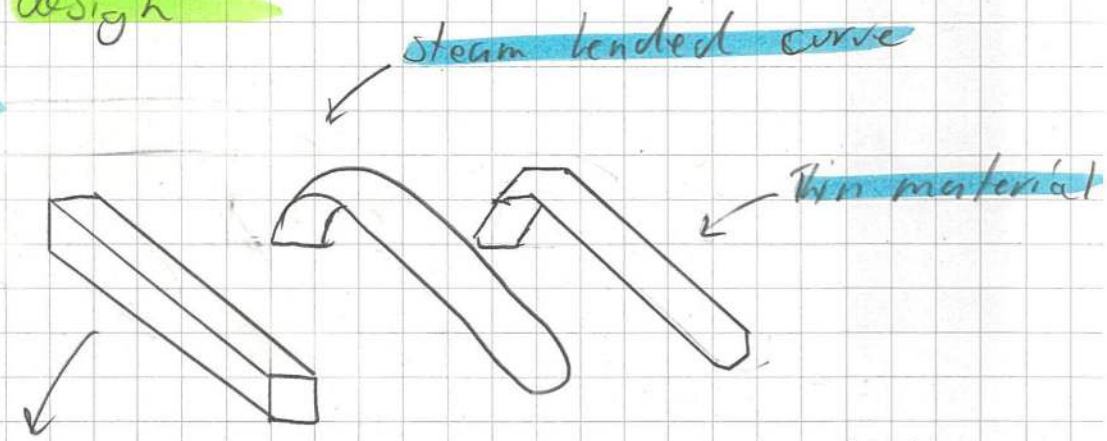
Supports under,

The end user would like the two seats to be connected but love the design

Arm Rests



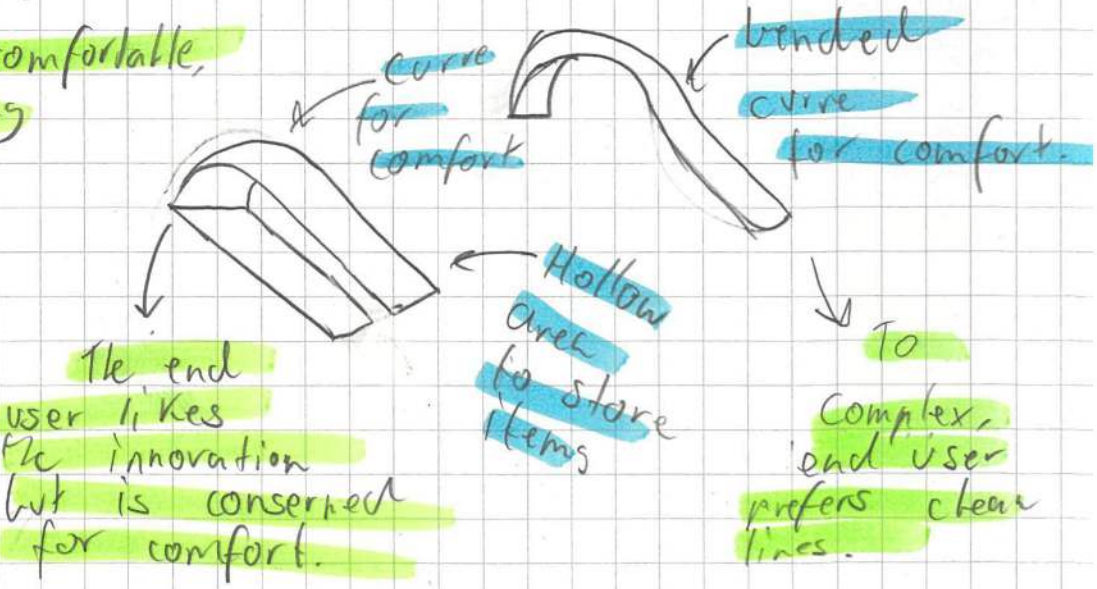
Arm rests



steam bended curve

Thin material

Not comfortable, is boring



curve for comfort

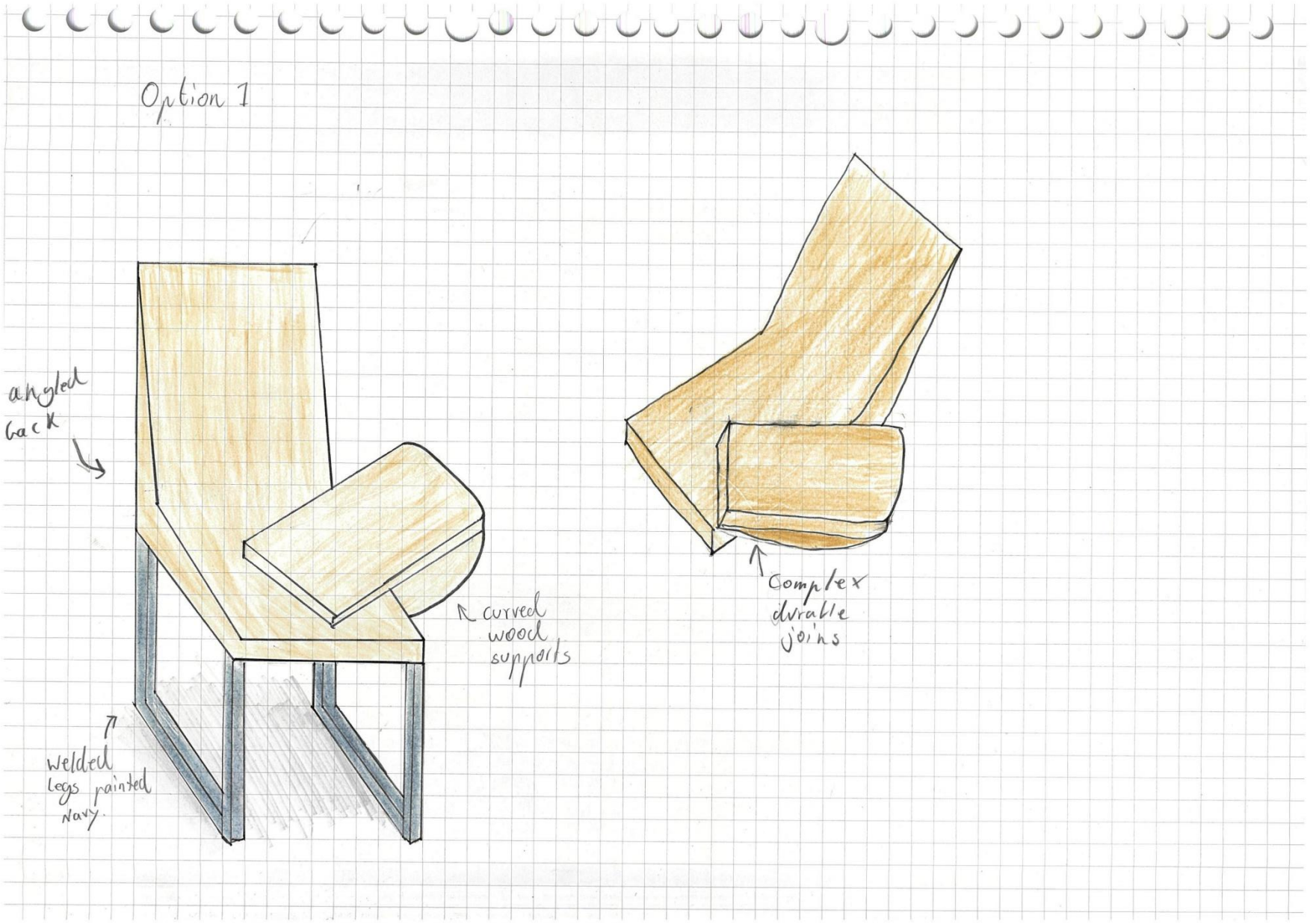
bended curve for comfort.

Hollow area to store items

The end user likes the innovation but is conserned for comfort.

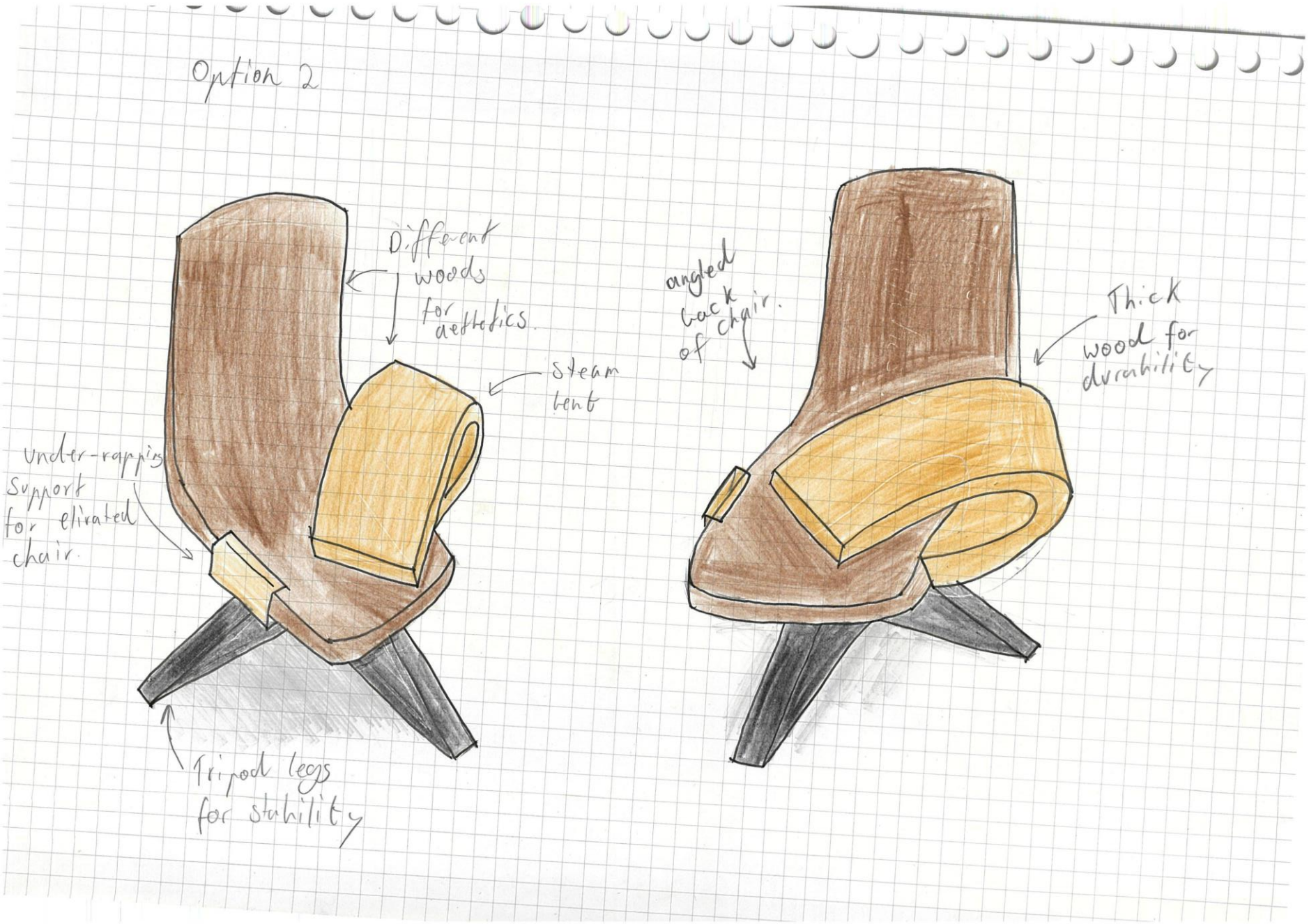
To complex, end user prefers clean lines.

Option 1



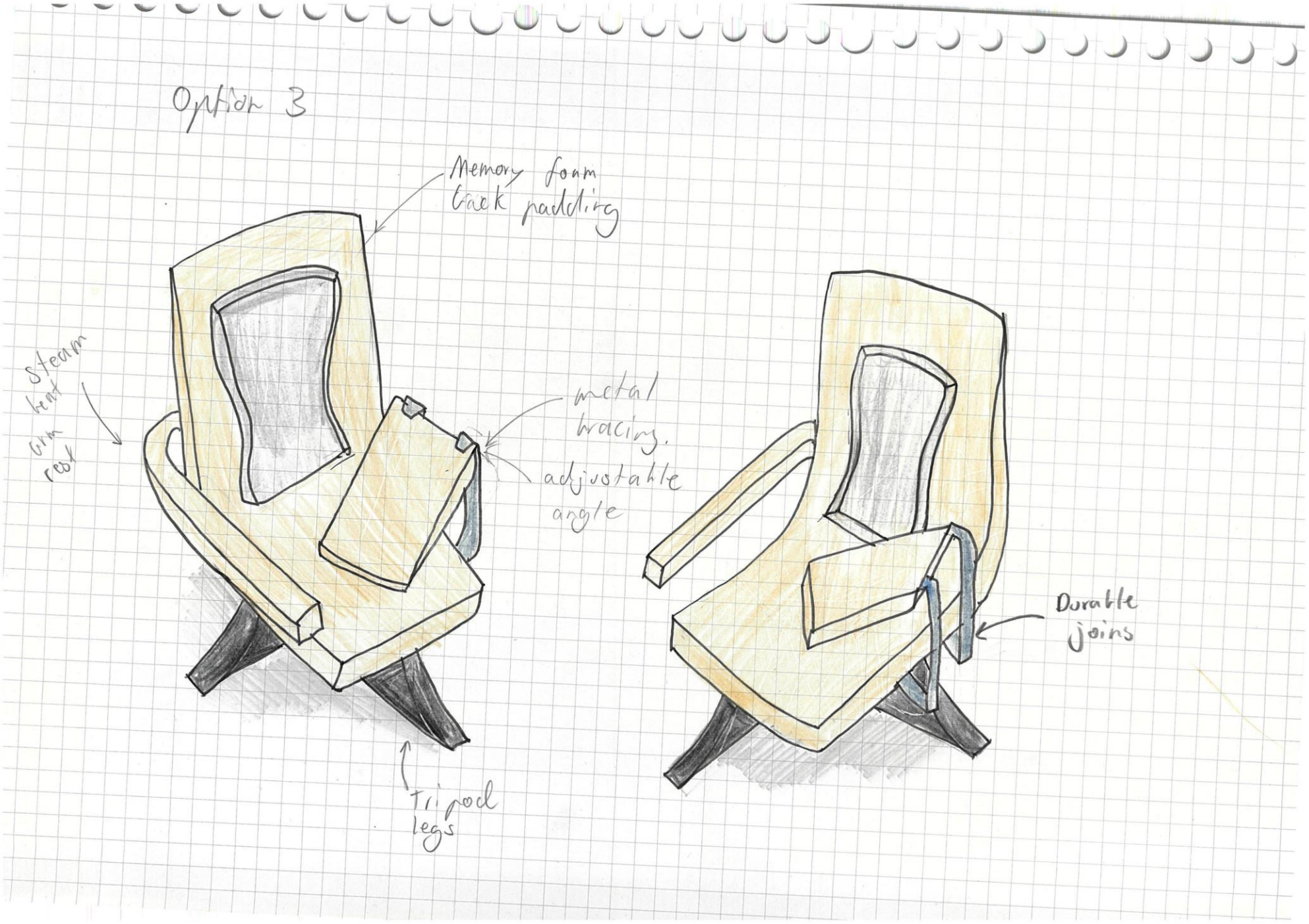
Criteria	Rating	End user feedback
Does the option account for Logan's expected future growth?	6/10	The end user has concerns that based on the design of this option it will not be able to support Logan's future growth. The main reason being the legs do not appear strong.
Is the option multifunctional and can be used without Logan?	6/10	The end user thinks that this option would be multifunctional as the elevated second seat could act as a table to place items on. However, the angle of the second seat could make this less than perfect.
Does the option appear safe?	2/10	The sharp edges of this option are a concern of the end user as they do not think it would be safe. Also, the legs do not appear safe for excess movement.
Does the option show signs of being durable and long lasting?	5/10	The end user has indicated that while the main form of the chair appears to be durable the legs and specifically the joinery of the legs do not appear to be long lasting.
Is the option easily accessible?	5/10	Due to the elevated seat being stationary the end user thinks that it would be difficult to get in and out of the chair. This limits the accessibility.
Does the option account for both end users' weight?	4/10	As the end user previously mentioned the legs of the chair do not appear to be able to withstand both end users' weights. The reasoning being the joins of the legs along with the thin material.
Does the end user find this option aesthetically pleasing?	2/10	No, the end user does not find the option to be aesthetically pleasing. They find it boring and lacking any unique design elements.

Option 2



Criteria	Rating	End user feedback
Does the option account for Logan's expected future growth?	7/10	The end user has indicated that the elevated chair appears sturdy and will account for Logan's future growth. However, as the elevated chair is stationary that could pose some issues with his growth.
Is the option multifunctional and can be used without Logan?	8/10	The ends user thinks that while the angle of the evaluated chair causes it to not be functional as a side table it could be used as an arm rest or to lean on,
Does the option appear safe?	6/10	This option does appear to be majority safe, the end user would like fewer sharp edges to improve the overall safety.
Does the option show signs of being durable and long lasting?	8/10	The thick materials and the joinery of the legs and elevated chair make this option appear durable, the end user is concerned about the strength of the steam bended curve as the angle of it seems to put stress on a certain point.
Is the option easily accessible?	10/10	The end user thinks that this option is accessible as having an opening on the side makes it easier to sit in.
Does the option account for both end users' weight?	8/10	Yes, the end user thinks that the tripod legs make the option appear to have the ability to hold the required weight and additionally account for any excess movement. The only improvement that could be made is the elevated chair have the less steep angle to remove the issue for stress on one point.
Does the end user find this option aesthetically pleasing?	9/10	The end user does find the clean lines and curves of this option to be very aesthetically pleasing. Some improvements would be having main form of the chair being the lighter wood instead of the elevated chair. And if possible, having exposed plywood laminations on the elevated chair as an added design feature.

Option 3



Criteria	Rating	End user feedback
Does the option account for Logan's expected future growth?	9/10	The end user believes that this option with the adjustable supports of the elevated chair defiantly accounts for logan's expected future growth.
Is the option multifunctional and can be used without Logan?	7/10	The end user notices that the adjustable function of the elevated chair makes it multifunctional as it could easily be a side table. The size of the elevated chair could pose and issue for how well it acts as a table.
Does the option appear safe?	5/10	The sharp edges of this option and the metal bracing for the elevated chair concern the end user as the effect the overall safety of the chair.
Does the option show signs of being durable and long lasting?	5/10	The moving hinges of the elevated chair pose an issue for durability as the end user is concerned they will break sooner than preferred. The end user thinks that the welded tripod legs are very durable and long lasting.
Is the option easily accessible?	7/10	The end user has indicated that the moving elevated chair make it easy to get into however the arm rest can overly restrict movements.
Does the option account for both end users' weight?	6/10	The end user is uncertain of the strength of the elevated chair given the moving parts. Having only two support beams does not appear durable. The main form of the chair appears very strong due to the legs and the wide base.
Does the end user find this option aesthetically pleasing?	6/10	The end user has issues with the lack on clean lines and thinks the armrest takes away from the overall balanced look of the option. The end user loves the idea of a cushion but would prefer it be recessed into the back of the chair and cover more area.

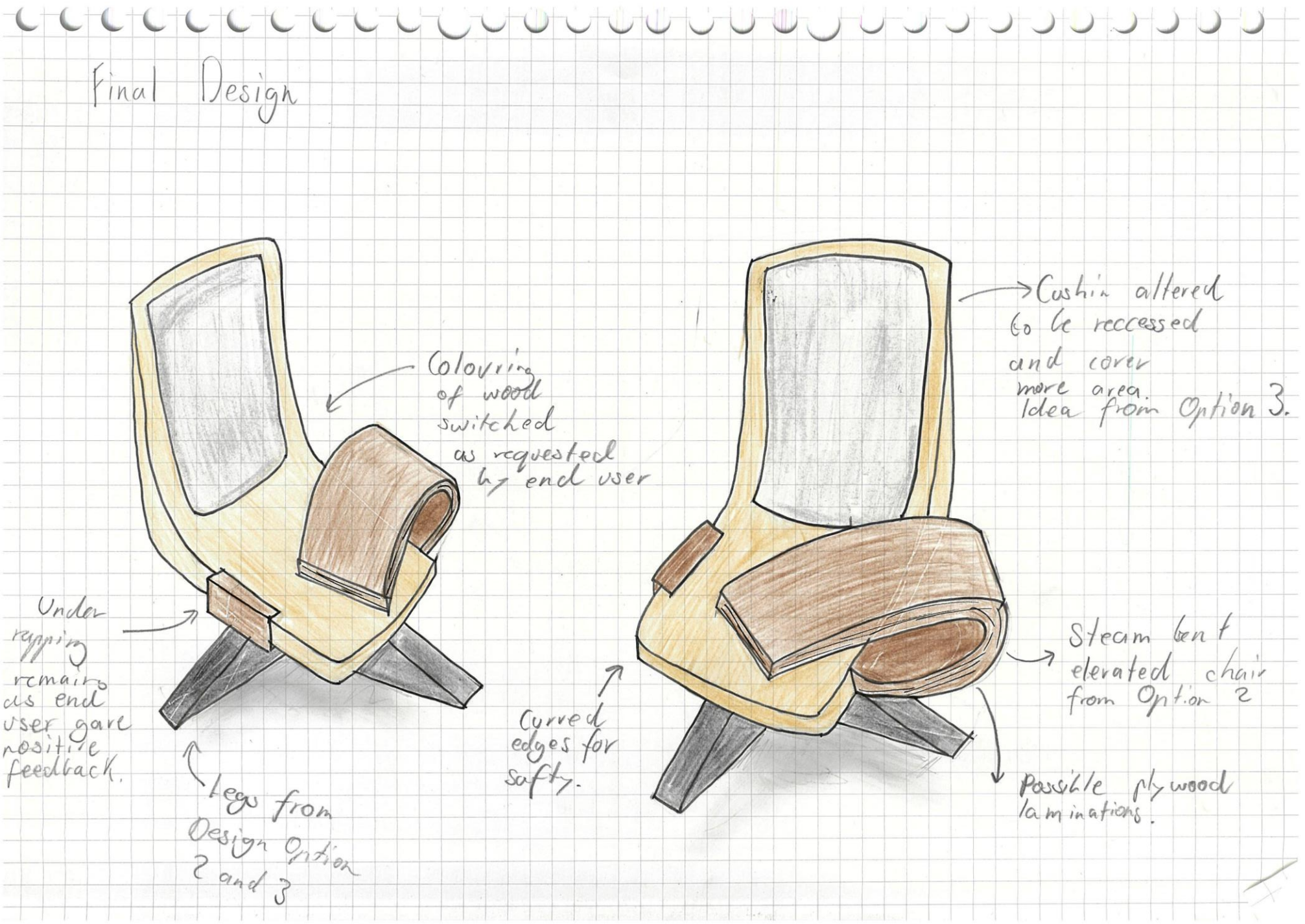
Preferred Design Option Justification

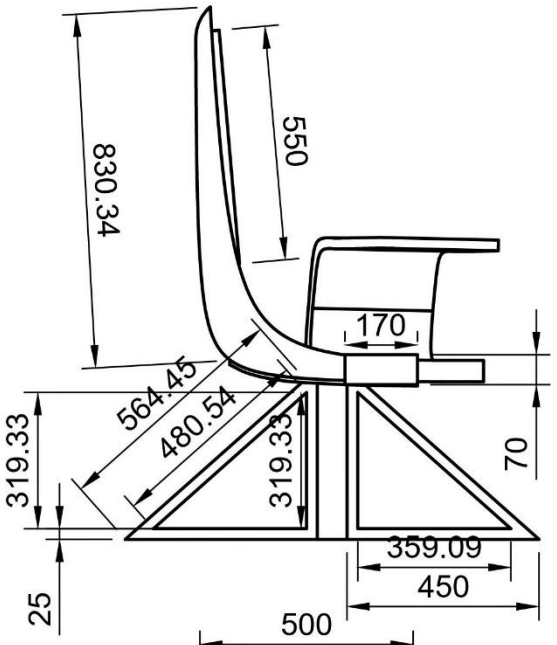
Design option	Total score
Option 1	30/70
Option 2	56/70
Option 3	45/70

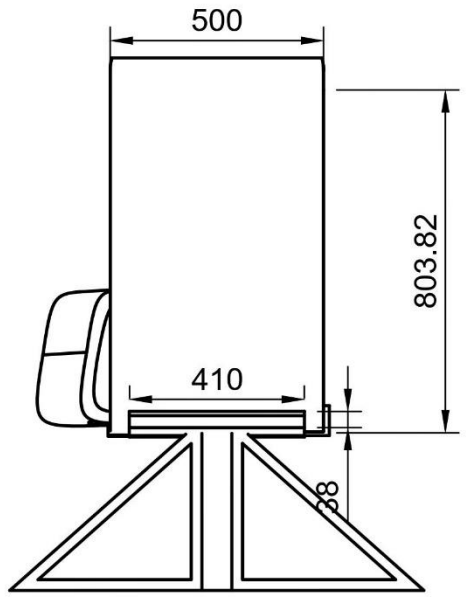
Overall, Melissa found things from each design that she would like to have included in the final design option, however based on the feedback given and the scoring she found Option 2 to be the closest to her desired final product. Melissa really likes how accessible the chair appeared and likes its clean lines and general aesthetic.

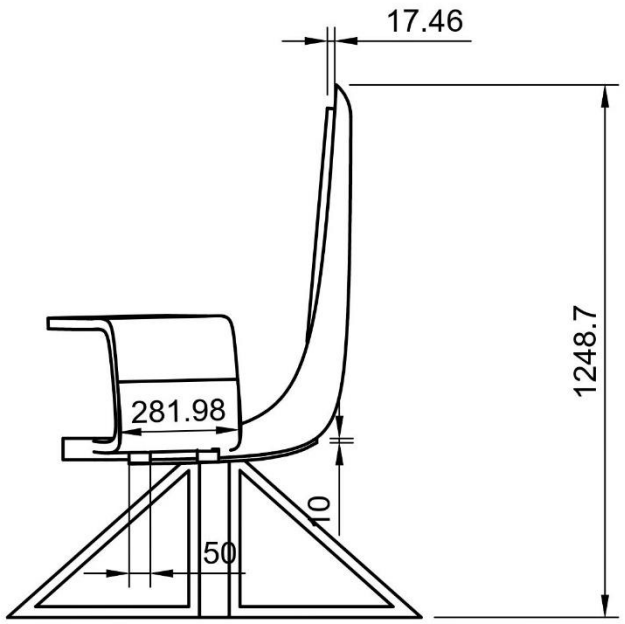
Improvements:

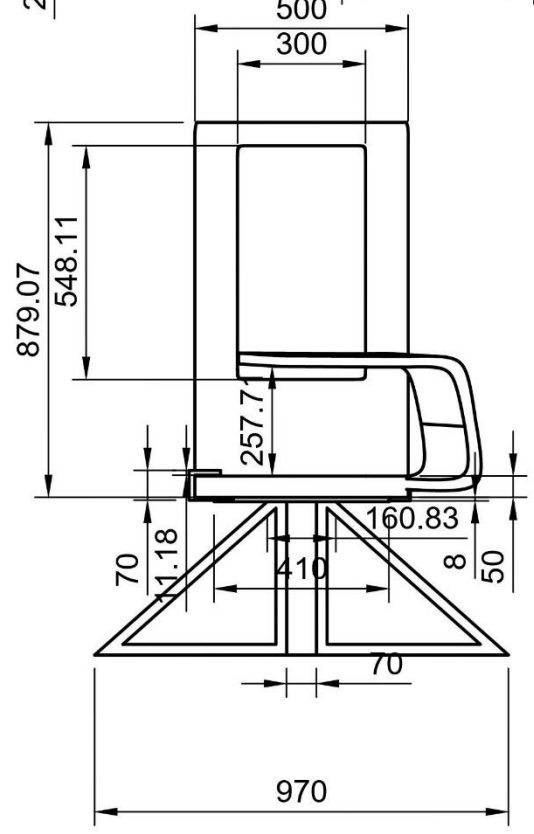
Melissa would love to incorporate the cushion into the final option but would like in to be recessed in the back of the chair and to cover more area. Melissa proposed that having the edges of the elevated chair have exposed plywood lamination or another form of adding a different design element, would greatly improve the aesthetic of the chair. Melissa stated that the size and angle of the elevated chair are important as they effect the durability and the multifunctionality of the chair. Increasing the size of the elevated chair would make it a more functional side table for when Logan is not using the chair. The angle was important to look at, due to the steep angle putting stress on a certain point. To improve the aesthetics of the chair melissa requested that the base chair form be made out of the lighter coloured wood and the evaluated chair be made from the darker coloured wood.

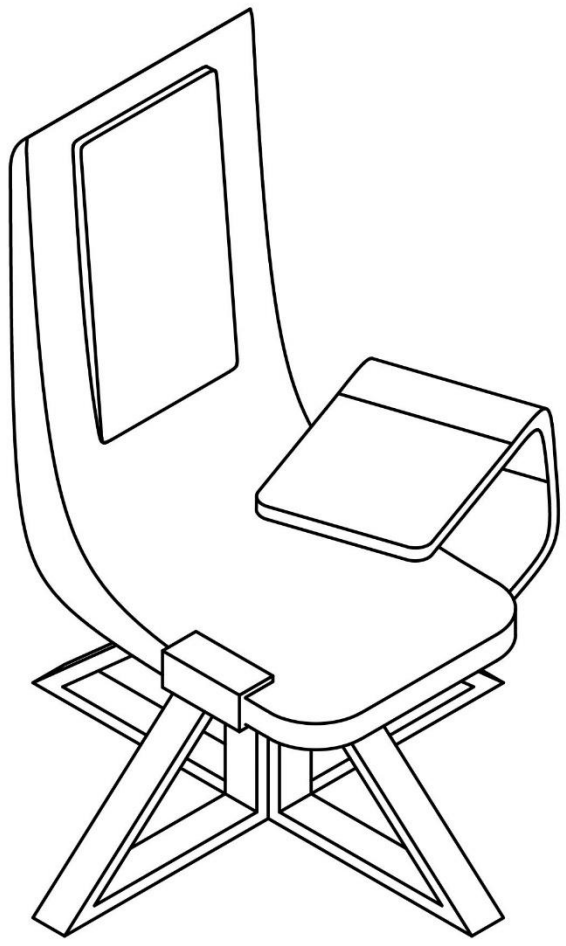




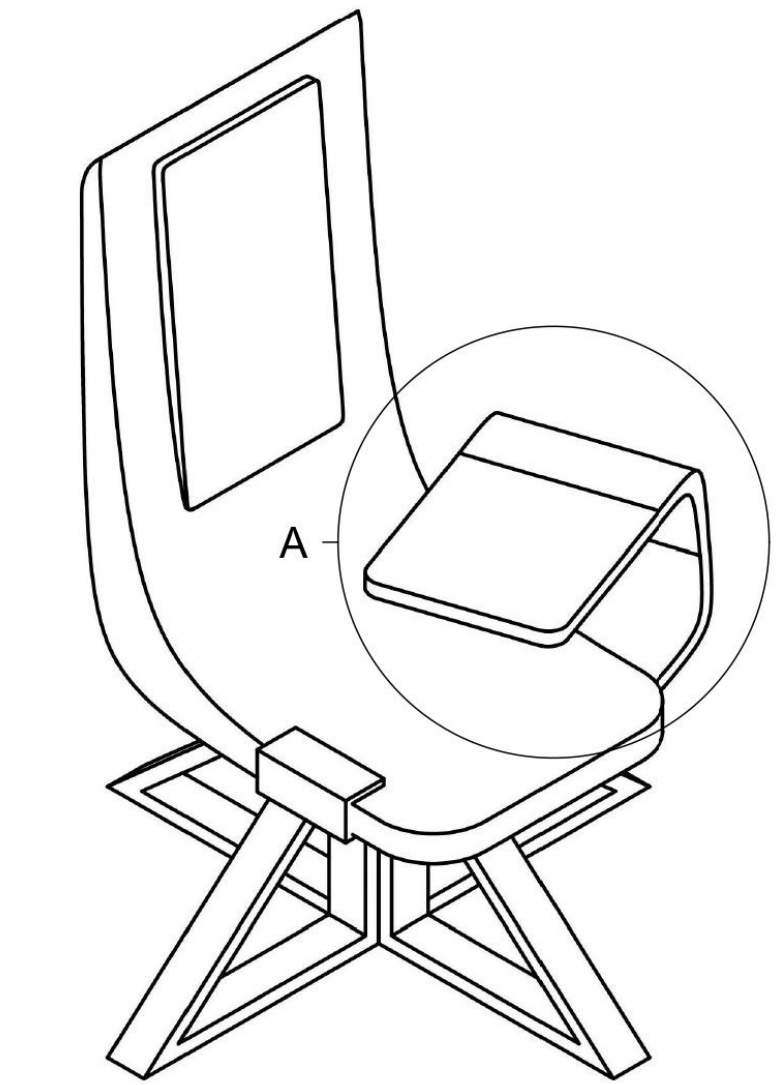




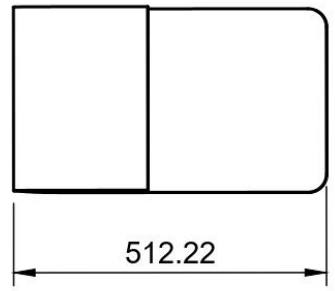


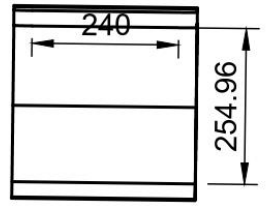


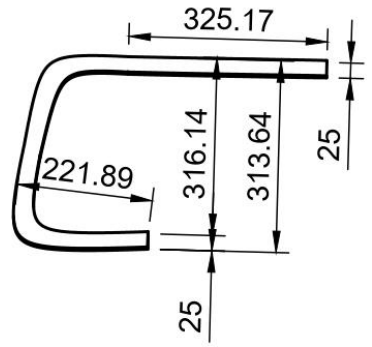
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		Isabella Handbury 22/04/2022	
1:15		Document type	Document status
		Title: Assistance Chair	DWG No.
			Rev. / Date of issue / Sheet: 1/1

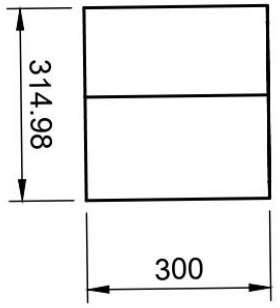


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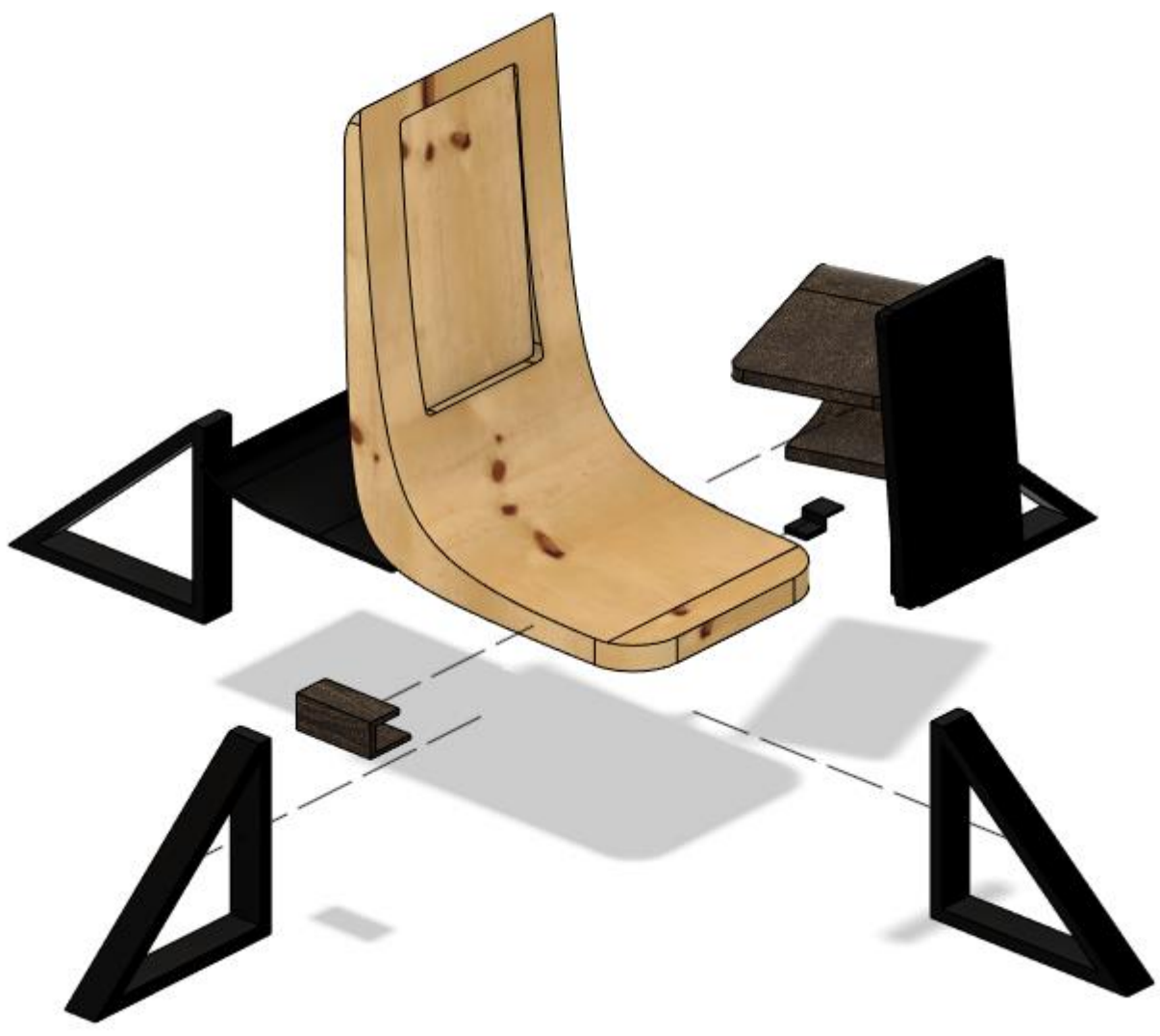




Dept.	Technical reference	Created by	Approved by
		Isabella Handbury 22/04/2022	
		Document type	Document status
		Title: Elevated chair	DWG No.
			Rev. / Date of issue / Sheet: 1/1

Model





Production plan

Step	Process	Materials	Equipment	Safety Precautions	Time Estimation (hour)
1. Measure and mark the steel	Using the working drawings as a guide, measure and mark out the dimensions on the steel. 450mm for the bottom pieces, 565mm for the angled pieces and 340mm for the vertical pieces.	Rectangular steel tubing	Marker Ruler Tape measure Set square	Handling of materials.	0:30
2. Cut the steel to the correct dimensions	Cut 4 pieces at 450mm, 4 pieces at 565mm and 4 pieces at 340mm.	Rectangular steel tubing	Hot saw	Wear the appropriate PPE including goggles, earmuffs and extra protective gear. Seek assistance in order to properly and safely use the tool.	1:00
3. Grind off any finishes on the steel and prepare it for welding	Using the angle grinder prepare the areas for welding. Using the previously made markings as a guide to where to grind.	Rectangular steel tubing	Angle grinder	Wear the appropriate PPE including extra clothing (long pants/jacket) to protect exposed skin.	0:30
4. Weld the steel into the triangle shape	Weld the steel into 4 right-angle triangles. The base being 450mm, the hypotenuse being 565mm and the vertical piece being 340mm. Use a set square to ensure the angle of the triangles is correct. Use the angle grinder to clean up the welds.	Rectangular steel tubing	MIG welder Angle grinder	Always wear the appropriate PPE including a welding helmet that is set to the correct specifications. Do not lift the helmet until the welder trigger is out of reach.	2:00
5. Grind the triangle legs until the top is level	Using the angle grinder ensure that the top of all the triangle legs is level and is ready to be attached to the base plate.	4x Steel chair legs	Angle grinder Level	Wear the correct PPE and cover exposed skin to protect from sparks.	0:30
6. Measure and mark the steel base plate	Accurately measure and mark out the dimensions for the steel base plate. 410mm by 430mm are the dimensions.	Steel flat bar	Marker Ruler Tape measure Set square	Handling of materials.	0:30
7. Cut the steel flat bar to the correct dimensions	Using the hot saw cut the steel into the dimensions of the base plate. 410mm by 430mm	Steel flat bar	Hot saw	Wear the appropriate PPE for the hot saw and seek assistance to safely and accurately use the tool.	0:20
8. Weld the triangle legs onto the base plate	Check with a ruler and set square before welding that the angle between the base plate and the 4 chair legs is correct and that when resting on the chair legs the base plate is level. Once the measurements are correct secure the pieces together and weld them together, getting all connecting surfaces. Clean up the welds with an angle grinder and test the strength of the welds.	Steel base plate 4x Steel chair legs	MIG welder Angle grinder Ruler Level Metal clamps	Wear the correct PPE for welding and griding and ensure that there is minimal exposed skin to protect from sparks. Check that the welding helmet has been charged and is on the correct settings and make sure it is always down when welding.	1:30
9. Measure and mark the posts of timber	Measure and mark the wood in accordance to the working drawings. Mark out the length of the wood to 500mm in addition to marking the measurements for milling the wood to the desired thickness. Each piece will vary due to the method of achieving the curved effect of the base chair. The posts will be cut like trapeziums on varying angles to gain the curved shape.	Cypress pine posts	Marker Rulers Measuring tape Set square	Handling of materials.	0:45

10. Cut the timber posts to the correct length	Using the table saw cut the timber posts to the correct length of 500mm.	Cypress pine posts	Table saw	Wear the correct PPE of safety goggles and earmuffs and keep hands clear of the saw blade.	0:30
11. Mill the timber to the correct thickness	In accordance to the markings, mill the timber posts down to the correct thickness. Each piece will have a varying thickness and shape in order to achieve the correct curve when joined. Refer to the specific drawings done for this step on page 46.	Cypress pine posts	Milling machine	Ensure a suitable extraction method is turned on before starting. Wear the correct PPE and keep hands clear of the blade.	1:00
12. Mark and align the timber for biscuit joining	Mark out and label the timber before cutting the biscuit slots. Label where each piece joins to another to ensure that they can easily be aligned.	Cypress pine posts	Marker ruler	Material handling.	0:30
13. Cut slots for biscuits	Using the biscuit joiner cut the slots in the timber for the biscuit joins. Using 3 biscuits along the length of each timber piece.	Cypress pine posts	Biscuit joiner	Wear the appropriate PPE for the tool and ensure that a suitable dust extraction method is in place.	0:45
14. Glue the wood together with the biscuits and clamp the pieces together	After the biscuit slots have been made, make sure that all the timber pieces align correctly with the biscuits in them then start gluing. Gradually glue each piece and clamp them together at the required angle to achieve the gradual curve.	Cypress pine posts Biscuits Wood glue	Clamps	Properly clean up the wood glue from hands.	1:00
15. Sand both faces of the chair to gain the desired curve	Using a combination of the belt sander and standard sandpaper of varying grits, sand both faces of the chair to make the curve more pronounced and to clean up the joined areas.	Cypress pine base chair	Belt sander Sandpaper	Keep hands clear from the belt sander and wear the correct PPE to protect from the dust and noise produced.	0:45
16. Sand the sharp corners and edges of the chair	Sand the edges of the chair down until they are dull and create curved corners of the chair base. Use a combination of sandpaper and the belt sander.	Cypress pine base chair	Belt sander Sandpaper	Keep hands clear from the belt sander and wear the correct PPE to protect form the dust and noise produced.	0:45
17. Mark out the placement of the recessed cushion	Measure and mark out the area that will be routed out to fit the chair cushion.	Cypress pine base chair	Marker Ruler Tape measure Set square	Material handling.	0:20
18. Use the router to create the area for the cushion	Clamp down the base chair then use the router and follow the markings to create the area that the cushion will be glued into. The dimensions are 300x548. The depth of the cut is 20mm.	Cypress pine base chair	Router Clamps	Take care with the direction the debris fly to and have a suitable extraction method. Wear the correct PPE of goggles and earmuffs.	1:00
19. Sand the edges of the recessed area	Use sandpaper to sand down the edges of the newly routed area. Starting with a low grit and moving upwards.	Cypress pine base chair	Sandpaper	Be cautious of the potential for splinter and dust from the sanding.	0:10
20. Attach the chair base to the legs	Drill 4 holes through the base plate that align with the chair base. Clamp the base of the chair into place and use the pedestal drill to drill the holes. The diameter of the holes should be the correct dimensions for a standard M8 bolt. Once the holes have been drilled bolt together the two pieces.	Cypress pine base chair Legs and attached base plate 4x M8 bolts 4x Washers 4x Hex nuts	Pedestal drill Clamps	Keep hands clear of all moving parts and make sure there is nothing that can cause entanglement with the tool. Wear goggles and earmuffs to protect ears and eyes.	1:30

Quality Measures

Quality Criteria	Measure	Importance	How to achieve?
Functionality	Ensuring that the final product is able to be used as intended. The chair's primary function is to hold two people comfortably, meaning it needs to be level, strong and allow room for future growth.	This is important as the product must be able to function as intended to satisfy the end user's needs. By operating as intended the product is able to be properly used by the end user.	Throughout the production stage the product will be continually checked to ensure that it is still working towards achieving the primary function. This includes checking the amount of weight the product can support, checking the level and testing the chair for comfort.
Durability	The product needs to last a long time, support the weight of two people, and accommodate for the end user's expected future growth. This requires the product to have durable materials, strong joins, and appropriate finish.	Constructing the end product to be durable with a long-life span is vital to the requirements of the end user but also means less money will be spend on replacement furniture and stops unnecessary waste.	Whilst selecting materials, the materials will be assessed to look for any warping, holes and the placement of knots and the grain will be checked to determine if they could affect the durability of the wood. All joinery will be done with extreme care to ensure that it is as strong as possible, and all cuts, welds and any construction processes will be done with as much accuracy as possible.
Finish	To ensure the product is completed to the highest quality a high-quality finish is required to be applied to all surfaces of the product. Additionally, the finish is needed to fit with the end user's aesthetics.	This is important as it not only effects the final look of the product but also the durability of certain surfaces.	Take care in sanding and applying the desired finish to the timber with the required number of coats. With painting check with the end user to finalise the colour before commencing painting.
Efficiency	Efficiency within the production stages is vital for being cost and time effective. This guarantees a high-quality product that is produced within the timeline previously established.	Efficiency allows more time to be allocated to the more important aspects of the production. Being efficient while producing the product saves time and money which is the best economical outcome.	This is achieved by using the most efficient equipment and production methods whilst producing the product. This must be done without cutting corners and sacrificing the quality of the product to save time.
Aesthetics	The overall look of the product and visual decisions made by the end user. Aesthetics not only relates to how the product looks but also its tactile qualities. The product needs to be produced to the aesthetic requirements put forward by the end user.	This is important as the end user must find the end product aesthetically pleasing if they want to place this product in their home and frequently use it. If the end user doesn't find the product visually or tactilely appealing other quality criteria could be affected.	This can be achieved by checking in with the end user repeatedly throughout the production process to determine if they agree with the decisions that are being made about the aesthetics of the product. The end user's feedback should be incorporated into the product.
Ergonomics	One of the main factors of this design is implementing appropriate ergonomic and anthropometric data to achieve the most comfortable final product for the end users. The product must be comfortable for the end users for them to use it frequently.	This is important as the design of the chair to hold two people requires lots of anthropometric data to be taken into account not only for comfort but also for durability and functionality. By implement the necessary data into the product it means that it will be easy to use and as comfortable as possible for the end users.	This will be achieved by double checking all measurements and referring back to the end users' anthropometric and ergonomic data. As well as regularly getting feedback from the end users about the ergonomic aspects of the chair and implementing their feedback.
Safety	Ensuring that the final product is safe for both end users as well as anyone walking around the final product. The safety of the product while its being used and while it's not needs to be considered.	It is absolutely vital for this product to be as safe as possible. Due to the way the product is designed with one person sitting above the other any accidents with the product could potentially seriously injure both end users.	This will be achieved by doing testing on the final product before giving it to the end user. Weight tests, movement tests and many other durability and functionality test will be completed to ensure the chair is as safe as possible.

Ghant Chart

Steps	Week 1			Week 2				Week 3			Week 4				Week 5	Week 6				Week 7		Week 8				Week 9		Week 10				Week 11		Week 12			
	9 may	11 may	13 may	16 may	18 may	19 may	20 may	23 may	25 may	27 may	30 jun	1 jun	2 jun	3 jun	6 jun	18 jul	20 jul	21 jul	22 jul	25 jul	27 jul	1 aug	3 aug	4 aug	5 aug	8 aug	10 aug	15 aug	17 aug	18 aug	19 aug	22 aug	24 aug	29 aug	31 aug	1 sep	2 Sep
1. Measure and mark steel																																				
2. Cut the steel																																				
3. Use the grinder to prepare the steel																																				
4. Weld the steel into the triangles																																				
5. Use the grinder to level the legs																																				
6. Measure and mark steel base plate																																				
7. Cut out the steel base plate																																				
8. Weld the legs to the base plate																																				
9. Measure and mark the timber posts																																				
10. Cut the timber posts																																				
11. Mill the timber																																				
12. Mark and align the timber for biscuit joining																																				
13. Cut slots for the biscuits																																				
14. Glue together and clamp the timber																																				
15. Sand the chair for the curve																																				
16. Sand the sharp corner and edges of chair																																				
17. Mark out the cushion area																																				
18. Use the router to create the area for the cushion																																				
19. Sand the edges of the recessed area																																				
20. Attach the chair body to the legs																																				
21. Measure and mark the elevated chair																																				
22. Cut the timber to prepare for laminating																																				
23. Create a jig to for the laminated timber																																				
24. Glue the timber and use the jig to clamp it together																																				
25. Measure and mark the steel supports																																				
26. Cut the steel for the supports																																				
27. Weld the steel for the supports																																				
28. Weld the steel supports onto the base plate																																				
29. Attach the elevated chair to the supports																																				
30. Measure and mark the Tassie Blackwood wrap around																																				
31. Cut out the timber for the wrap around																																				
32. Glue the wrap around together																																				
33. Attach the wrap around to the chair																																				
34. Sand the chair																																				
35. Varnish the chair																																				
36. Paint all steel components																																				
37. Glue in the cushion																																				
38. Extra time to allow for missed production lessons and to allot for production methods taking longer than expected																																				

Risk Assessment

Equipment, tools, machines	Type of possible hazard	Potential injury	Level of risk		Safety Precautions
			Likelihood (l, m, h)	Severity (l, m, h)	
Welding	-Overexposure to welding fumes -Bright light from arc -Heat from the arc -Fire and explosions	-Damage to lungs from fumes -Blindness -Burns -Range of severe injuries due to explosions/fire. (eg. Burns, shrapnel)	L H M L	H H M H	Make sure the area of welding is properly ventilated and is a designated area for welding to protect other students. Wear a welding mask, and ensure it is on before, during and after then welder is being used to prevent blindness. Make sure the work area is clear and clean of any flammable materials. Know the location of fire alarms and extinguishers.
Drop saw	-Exposed moving parts -Moving saw blade	-Risk of danger to hands from the saw blade -Eye injury from saw dust	M M	H M	Ensure that hands are clear of saw blade before using the tool. Wear safety glasses to avoid injury from saw dust.
Angle grinder	-Kickback -Equipment breakage -Moving saw blade -Sparks	-Contact with projectiles (eg. broken saw, material pieces) -Sparks contacting the skin or eyes burning them.	L H	H M	Wear the appropriate safety gear to protect against possible injury from projectiles and sparks. Angle the saw correctly to direct the flow of sparks away from the user.
Sanding	-Dust in eyes -Inhaling dust -Cuts and abrasions to hands -Excess loud noise	-Eye damage (temporary or extended time) -Lung damage due to the inhalation of dust -Damage to hands from contacting the sanding disk -Damage to hearing from continuous loud noise	M L M L	M M L M	Wear protective eye ger to avoid dust damaging eyes. Ensure that a dust extractor is being used if possible in combination with wearing a dust mask to minimise the risk of inhaling dust. Ensure hands are kept a safe distance from the sander. Wear earmuffs to protect ears from the continuous loud noise.
Biscuit joiner	-Sharp blade -Excessive dust -Excessive noise	-Risk of injury to hands due to the sharp blade -Injury to eyes and lungs from the dust -Damage to ears from the continuous loud noise	M M L	M M M	Wear safety goggles along with earmuffs to protect from the excess dust. Make sure a dust extraction method is also in place to help minimise the risk. Keep hands clear of the blade area whilst using the tool.
Band saw	-Moving saw blade -Dust in eyes	-Risk of injury to hands from the saw blade -Eye injury from dust in eye	M M	M M	Make sure that hands are a safe distance from the blade and care is taken when cutting smaller materials. Wear safety glasses to protect eyes from dust.
Table saw	-Moving saw blade -Dust in eyes -Excess loud noise	-Risk of injury to hands from the saw blade -Eye injury from dust in eye - Damage to hearing from continuous loud noise	L L L	M M M	Keep hands clear of the blade and don't clear and small cut off pieces while the blade is moving. Wear safety glasses to protect eyes from dust. Wear earmuff to protect ears from continuous loud noise.

Industry Manufacturing

Manufacturing:

Due to the nature of the product and the complexity of production in industry, the product would be produced in low volume. Another aspect that would factor into the scale of manufacturing would be researching the market and determining if there is a greater need in the market for the product. The product would need to be altered to use standard ergonomic data instead of the anthropometric data collected from the end user as ergonomics would better suit a large group of people.

While various machinery would need to be used at industry level production for the most cost effective and time efficient manufacturing, lots of the production would still be hands on work to ensure the quality is high. The chair being handmade could have the potential for a market edge due to the higher quality product being produced.

The product would be strictly made to order because of the high cost to produce the chairs made to order ensures that there is a consumer that wants to buy the product before it is made. Even though the consumer wait time would be higher inventory would be easier to control and there wouldn't be as much dependence on market demand.

Another reason for the chairs being made to order is that it would allow for a highly customisable end product that suits the needs of each end user. Not only would the chair be custom is terms of anthropometrics but could also cater to the needs of the end user's individual disability. An example of this would be certain textures or finishes of the chair would fit better with one end user's sensory needs than others.

Market Research:

For this product to be manufacture at industry level more market research would need to be carried out with Autism community to gain information and feedback to alter the chair so it is more appealing for the market. Primary research would be gathered by attaining data from surveys as to reach a broader group, but the main information would be gathered from field trails, with protypes being tested and discussions being had about the product.

Research and Development:

Some research could be put into developing a new hybrid material that is easy to steam bend whilst still being as durably as possible. While this would take time and cost money, the long-term benefits would be a higher quality product, easier manufacturing, and less wastage from failed steam bending attempts.

The best ways to achieve lean manufacturing with this product could be researched. This would ensure that the least amount of waste possible is produced. This is beneficial to both the cost of materials but also the environment.

Research would be done into determining the best machinery to produce the assistance chair along with developing better machinery to steam bend the elevated chair to the correct specifications in the most economical way possible. This along with research into new and emerging technologies would ensure that the manufacturing of this product would be as sustainable, economical and produce the highest quality product possible.

Machinery:

A Laser cutter is a possible piece of machinery that at industry level would prove to be an effective tool for making a consistent product. By using the laser cutter, each cut would be as accurate as possible and would cut down on wastage due to minimal errors. While laser cutting machines include a wide spectrum of potential uses in a variety of industries, they are becoming increasingly useful for designers, artists, manufacturers and schools. The laser cutter would be efficient, reliable and accurate and would help to produce a high-quality product at industry level.

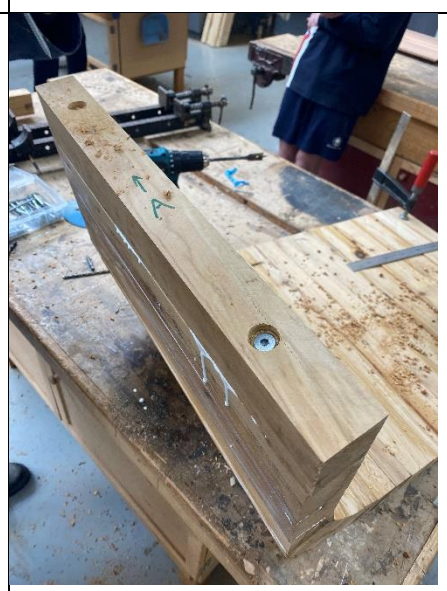
A CNC router would be a very useful machine to have at industry level due to its computer controlled cutting process, which is more accurate, faster and safer than a handheld router. The CNC router would be perfect for routing out the area of the chair where the cushion would go.

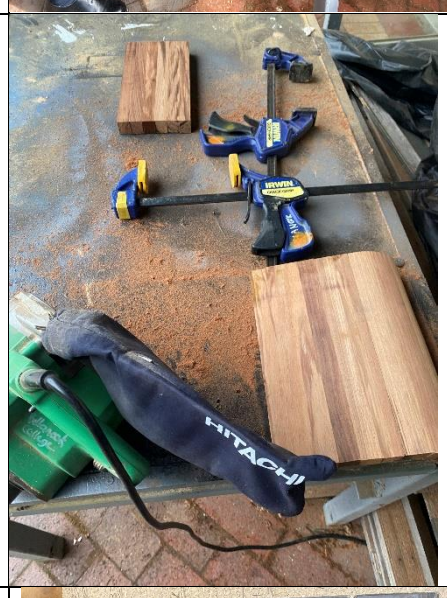
A CNC milling machine which accurately cuts down the wood to the correct thickness according to the specifications that have been designed on the computer would increase the speed and efficiency of producing the product. The CNC milling machine would cut down on the wood required to make the process of joining the timber planks for the base chair easier and less time consuming.

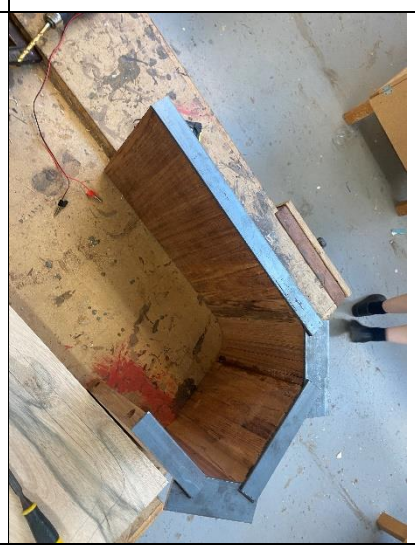
When producing the product in a larger scale instead of a one off, the construction of a jig or multiple jigs would be reasonable for producing the elevated chair in conjunction with steam bending it. The use of a large-scale steam box would be the perfect way to create the bent shape whilst still having a strong elevated chair. The steam box would allow the wood to become pliable enough to be fitted into the custom jigs to cool and harden into the desired shape

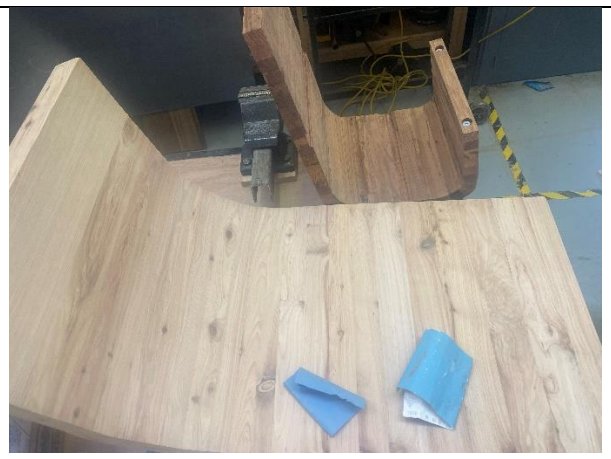
Production Journal

	Significant additional steps

17. Align the pieces of pine and use wood glue to laminate them together		This process was done for the pieces of pine that were on the flat edge as clamping at this angle worked well. This type of joining was also done as this piece of the chair doesn't require as much strength in the joins.		0:30	The quality method of checking the levels of the pieces of wood before finalising the gluing made sure that the back of the chair would look aligned and be stronger.
18. Line up the curved pieces and use wood glue and 100mm long screws to join them together		The curved pieces were lined up to the now glued back of the chair and pilot holes were drilled for the screws. Once the pilot holes were lined up glue was added to both sides of the soon to be joined pieces. While holding the pieces in place an impact driver was used to screw in the screws. This was repeated for each individual piece of wood. An addition to the modifications log was added due to mistakes made with this step. The mistake was that despite measuring and drilling the pilot holes the ends of some of the screws came through the back of the chair and chipped some of the wood.		0:30	Pilot holes being drilled to help align the pieces of wood along with ensuring the angles were correct, were excellent quality measures in theory however they were ineffective and the mistake of the screw going through the back was still made. The following additional steps will have quality measures to ensure that this mistake does not effect the quality of the chair.

19. Sand both faces of the chair to gain the desired curve	Using a combination of the belt sander and standard sandpaper of varying grits, sand both faces of the chair to make the curve more pronounced and to clean up the joined areas.	The only minor modification made to this step of production was to use handheld sandpaper to even out the internal concave curve of the chair. This was required as the bend sander was unable to properly reach the angle due to the shape of the sander's frame.		0:30	To ensure that the final finish of the sanding was as smooth as possible the belt sander was only used with the grain not against it. To adhere to safety measures all appropriate PPE was worn. This included apron, earmuffs, mask and goggles.
20. Sand the sharp corners and edges of the chair	Sand the edges of the chair down until they are dull and create curved corners of the chair base. Use a combination of sandpaper and the belt sander.	Due to the large percentage of the corners that needed to be curved including both the elevated and base chair various tools including the chisel were used to take off some of the corners.		0:30	The sand paper was used in various grits to achieve the highest quality finish. All safety gear was worn to minimise risk.
21. Measure and mark the steel base plate	Accurately measure and mark out the dimensions for the steel base plate. 410mm by 430mm are the dimensions.	An alteration to this step in the production process was to change the dimensions of the base plate to better accommodate the size of the main body of the chair. The sizing was also changed to better allow for any components that needed to be attached in this case the supports for the elevated chair. Another reason for the change was to account for fixings that would be required to secure all components together.		0:20	The quality measures taken for this step of production were to use tools to ensure all measurements were as accurate as possible. This included using set squares and a variety of different length rulers.
22. Cut the steel flat bar to the correct dimensions	Using the hot saw cut the steel into the dimensions of the base plate. 410mm by 430mm	Due to the change in the size of the markings (see above) the cutting sizes were also altered. The new measurements that account for the change in size requirements are roughly 50mm x 450mm with each piece having a slightly different length and angle to		0:45	When cutting the steel flat bar all safety precautions were taken to minimise risk. This included keeping the area surrounding the saw clean and other students being alerted to the saw being in use.

		leave room under the main chair to mount the elevated chair. this means that additional welding is required for the next step to weld the 4 different pieces of flat bar together.			
23. Weld the triangle legs onto the base plate	Check with a ruler and set square before welding that the angle between the base plate and the 4 chair legs is correct and that when resting on the chair legs the base plate is level. Once the measurements are correct secure the pieces together and weld them together, getting all connecting surfaces. Clean up the welds with an angle grinder and test the strength of the welds.	As mentioned in the previous step additional welding was required to connect the 4 different strips of steel flat bar.		1:00	When welding all proper safety precautions were taken and all correct PPE was worn. Including welding mask, welding gloves, welding jacket and long pants. The welding was completed in a designated space to keep others safe as well.
24. Measure and mark steel for the elevated chair supports	Measure and mark out on the steel flat bar the dimensions for the supports for the elevated chair. Mark out a width of 50mm and varying lengths.	The dimension of the elevated chair supports were altered to better align with the produced elevated chair not the design. This better accounts for the curve of the elevated chair and the various different thickness of the curve.		0:40	To ensure the measurements were as accurate as possible, a variety of measuring tools were used and measurements were double checked.

29. Sand the wooden portions of the chair that are yet to be sanded	Set up the chair so that it is easy to access and so it is secure. Using a combination of grit sandpaper to start the off process of achieving a high-quality finish on the chair. Sand the wrap around detail and the elevated chair starting off with 80 grit and moving up to 120 grit.	The only notable change to this step was also using the belts sander and chisel to take off larger portions of wood.		0:50	To achieve the highest quality finishes time and care were taken when sanding the chair to prepare for varnishing. As mentioned in the intended process this included the transition from high grit to finer grit sandpaper.
30. Varnish the wooden section of the chair	Using wood varnish evenly distribute the varnish to the wooden section of the chair. Apply the varnish many times to achieve and multi layered smooth finish.	No modifications were made.	 	1:10	Many light coats of varnish were applied to create the best outcome with the texture and appearance of the chair.
31. Paint the legs of the chair and the base plate and any steel supports.	Paint the steel supports, chair legs and steel base plate with a matte black paint. Apply multiple coats of the paint to achieve and high-quality finish.	The only alteration made this step was to add primer before adding the black paint.		1:30	To achieve the best quality finish primer was added before the painting to improve the quality of the end result.

Ghant Chart

Actual:

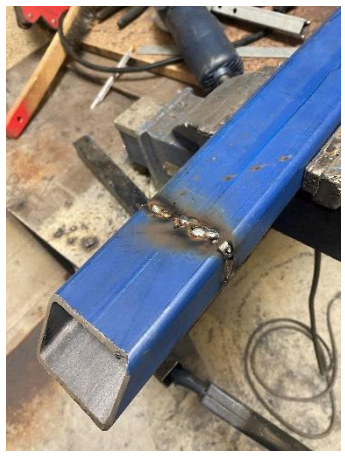
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1. Measure and mark the steel																																				
2. Cut the steel to the correct dimensions																																				
3. Grind off any finishes on the steel and prepare it for welding																																				
4. Weld the steel into the triangle shape																																				
5. Apply builders Bog to the holes in the welds then sand it back to create the desired effect																																				
6. Clamp the pieces together at right angles to prepare for welding																																				
7. Weld two pieces together at 90-degree right angles. Repeat for both pairs of legs																																				
8. Align the two pieces to weld all the legs together and weld them to have the finished leg form																																				
9. Measure and mark the dimensions for the elevated chair																																				
10. Cut the timber to the correct thickness and prepare it for laminating																																				
11. Mark out the angles for the curved pieces and cut them with the table saw on the desired angles																																				
12. Align the pieces and check for any inconsistencies with angles																																				
13. Glue the pieces together and hold them together with a sash clamp																																				
14. Measure and mark the posts of timber																																				
15. Cut the timber posts to the correct length																																				
16. Mill the timber to the correct thickness																																				
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32. Extra time to allow for missed production lessons and to allot for production methods taking longer than expected																																				

Significant additional steps:	

Modifications Log

Throughout the production stage it generally uncommon for every process to go as initially planned. However, it is very common for modifications to be made to correct a mistake or after a realisation is made on ways to improve the final quality of the product. Even though my planning process was relatively thorough, many modifications were made due to unforeseen errors and to improve quality or speed up the production process in order to focus more on the more important factors of my end product.

Reattaching a miss cut steel leg:



In the early stages of production errors were made when calculating and measuring the lengths of the different sides of the triangles. Luckily, this mistake was quickly discovered and only resulted in one of the lengths needing to be fixed. The piece was cut too short and the limited amount of steel that was ordered inhibited me from recutting a new length. To combat this issue a small offcut from one of the other lengths was cut to the required size with the angle on one edge. Both pieces were then prepared for welding including grinding back any burrs and the paint. They were then aligned with set squares and then placed in a clamp. Once 3 of the 4 sides were tack welded the piece was flipped and the inaccessible side was welded. Once all the sides were welded the excess weld was grinded back with the angle grinder and any holes discovered by grinding back the weld were then filled and the process was repeated until there was a clean weld. This error ended up having no effect on the final product due to the care that was taken when fixing the issue, resulting in a successful modification.

Oxy torching the legs back into 90 degrees:



When welding two of the triangles together at 90-degrees, heat warp was not accounted for when running a long line of weld. This resulted in the large amount of weld causing the pieces to warp out of 90-degrees and becoming an acute angle. This error could have been a major setback due to having all four legs in square being an integral part of the stability of the chair. This mistake was combatted by taking an oxy torch to the long line of weld and using a sash clamp to slowly bend the pieces back into square. The use of the oxy torch resulted in some discoloration on the metal, but this wasn't an issue due to it being hidden by the enclosed shape the four legs make. When welding the other two legs together many measures were put in place to ensure that the same mistake didn't occur again. These measures included using more clamps to hold the pieces in square whilst welding and taking more time to use a set square to make sure the pieces were square. The main change to inhibit the same mistake from occurring was in the technique of welding. Firstly, the two pieces started slightly out of square the when the tack weld was applied then warped into square. The other change that was made to the welding technique was to run 4 x 50mm beads of welds to reduce the heat warp of the chair legs. These methods of fixing the error resulted in a successful modification, where the error is not noticeable and doesn't affect the final quality of the product. (the image shows the difference in the welds. Bottom is the oxy torched modification. Top is the second attempt with a better technique)

Method of curving the elevated chair:



The plan for this stage of production was initially to create a jig to laminate the wood by cutting thin strips of wood, soaking them, then gluing them in the jig with the curved shape. This process was modified to save time and to produce a higher quality product. The modified process was to have the Tasmanian blackwood cut in 500mm lengths with a variety of widths. Using the sketch seen in the photo the pieces were aligned and grouped by their colour to create a quality aesthetic product. The thicker pieces of the Tasmanian blackwood were used on the curved corners to create more stability. The angles were then calculated and marked out on the wood. The desired angles were then cut on the table saw with the use of a digital angle finder to adjust the saw to the required angle of each piece. When all the angles were cut the pieces were aligned then glued together. To achieve the smooth curve the excess wood was removed by running the pieces though the bandsaw then by using the belt sander the smooth curve was achieved. This modification was made as it was predicted to result in a higher quality and more durable elevated chair which is vital as it has the function of supporting a considerable amount of weight. This prediction was correct and the modification also resulted in saving time due to the reduction of complex steps.

Method of creating the curve of the main chair and adding internal bolts:

The initial plan for this stage of production remains predominantly unchanged however small modifications were made to improve the quality of the final product. To create the varying thickness of wood for the curve the plan was to mill down the wood to the different dimensions. During production the realisation was made that this process would be less accurate and more time consuming then running the pieces of wood through the table saw set to different angles and widths. These angles allowed the pieces to fit together smoothly to create the curve.





To add additional strength and support to the main chair large bolts were drilled through the wood. This process included measuring and drill pilot holes to attempt to stop the screws from breaking through to the other side. The use of a large drill bit to create a hole for the head of the screw to rest in to prevent it from causing issue with the next piece. After these steps where complete, wood glue was spread on the face of the wood and the screws were secured with an impact driver. With securing the final piece of wood no screw was put in as it would have had to be exposed at the front of the chair, this creates no issue for the strength of the main chair as that piece is not load bearing. This modification process helped to produce a more durable product that better fits the end user's requirements for strength in the final product.

Fixing the problem of the bolts going through:

In the modification process seen above (method of creating the curve of the main chair and adding internal bolts), an error was made when measuring the angle of the pilot holes for the large screws. This error led to the screws piercing though the back of the chair and chipping some of the wood. While at the time this mistake seemed to be a major set back it was easily fixed. To combat the issue firstly the ends of the screws were removed with the angle grinder, the next stage was to reattach the chipped pieces of wood back to the main chair. This step was completed with applying wood filler to the problem area and once dry sanding it back. This solution worked remarkably well and the mistake is unnoticeable. While completing the process of adding the screws always posed the risk of this error the step was necessary to create additional durability of the product, so it was imperative to have a solution to the problem.



Removing the cushion from design:

The modification of removing the cushion from the chair was completed after numerous conversations with the end user regarding the aesthetic properties of the chair. The end user decided that they found the clean lines and simplistic properties of the wooden main chair very aesthetically pleasing and disliked the idea of taking away from that by adding the cushion. The end user felt that by adding the cushion the quality of the chair would be reduced, and that they would end up liking the appearance of the chair more without the cushion. Another reason proposed by the end user for why they wanted the cushion to be removed from the final product was that by having a fixed glued cushion, the chair wouldn't be adaptable to her lifestyle and that it was impractical for washing. Being able to clean the cushion was a must for the end user due to the purpose of the chair being a stable everyday piece of furniture for her and her son, who she mentioned can be messy at times. After conversing with the end user, the final decision was made to remove the cushion from the chair due to aesthetic and practicality reasons.

Removing the wrap around piece from product:

After contemplating the design of the chair with and without the wrap around piece the end user came back with a number of reasons why they would like it removed from the final product. The initial one being that when they tested getting in and out of the chair they realised that the wrap around piece would restrict their movement and ease of getting out of the chair. Another reason was that they felt it distracted from the key aspects of the chair and that the sharp edges greatly contrasted the smooth curves of the chair. After conversing with the end user, the decision was made to remove the piece from the chair design.

Adding steel supports around the outside of the elevated chair:

The addition of the steel supports around the outside of the elevated chair was made for two reasons. The first was to add much need strength to the elevated chair, the second was to enhance the appearance of the chair. The realisation that the initially planned method of having steel supports under the elevated chair would not be strong enough to support the end user's weight created the need for an innovation solution to a problem effecting a crucial part of the product. To solve this problem the course of action was to create steel supports that follow the curved line of the elevated chair and created a much-needed rigidity to the weak points of the elevated chair. These additional supports connected to the steel base plate of the chair which put the strain on the legs of the chair which are the strongest part. The change to this style of supports also enhanced the appearance of the chair. The painted steel matched the legs of the chair which tied the entire product together and added some much-needed cohesion. This modification was crucial to producing a product that the end users could safely use and had the added bonus on enhancing the aesthetics of the end product.



End User Instructions

The materials, construction methods and overall design of my product creates a final product that is best suited for the end user. However, for the product to be used to its best ability instructions need to be followed. The main factors that the end user needs to consider when using the chair are weight and materials both for safety and for the proper maintenance of the product.

General care

For general maintenance and cleaning of the chair dust the surfaces with a soft cloth and if spillages occur immediately wipe them away with a damp cloth. To avoid discolouration of the wood refrain from placing the chair in direct sunlight for extended periods of time. Due to the properties of the steel legs, it is recommended that if the chair is intended to be placed on a wooden surface that floor protecting felt pads are added to the legs of the chair alternatively the chair could be placed on a rug. This is to prevent damage to the wooden floor surrounding the chair.

Wood maintenance

Due to my product being finished with a varnish, careful care instructions need to be followed to keep the wood in the best condition possible. The basic instructions for regular upkeep of the wood include; protecting the varnished wood from water and other liquids and regularly cleaning with a dry cloth. In the case of spilling anything on the timber or to clean off any dirt use a damp cloth followed by a microfibre towel to dry the surface.

Due to the multifunctional nature of the chair, drinks can be rested on the elevated chair section. While it is recommended that a coaster be used with all drinks there is a method of removing water rings should they occur. The best method of removing water rings on varnish is to start by applying an equal parts olive oil and vinegar mixture to the affected area and with a cloth and wiping it in the direction of the wood grain, until the stain is gone. This the stain doesn't disappear a more abrasive mixture of non-gel toothpaste and baking soda can be used with the same method to try to remove the stain. Both mixtures need to be properly cleaned off the wood with a damp cloth then dried with a microfibre towel.

Over an extended period of time the varnished wood may need to be more thoroughly restored. To achieve this the varnished wood can be protected and restored by applying a coat of paste wax. The wax paste protects the wood from moisture and dust. This does not need to be applied often as excessive application of the paste wax will cause build up. The minimum period of time between wax applications is one year however much longer gap between applications is recommended.

Steel maintance

While the steel portion of the chair requires little to no maintenance other than occasional dusting, the painted material is subject to scratching. To fix any scratches that may occur over time with the steel on the chair a mark up paint pen is recommended. A mark up pen can easily fix any scratches and leaves the product with not noticeable markings.

Transport

Because of the obscure shape of the chair and the weight of the final product it is not suitable for a single person to lift. A minimum of two people should be helping to move this chair and they should be using the proper lifting techniques for heavy lifting.

Steps for proper lifting technique.

1. Keep a wide base of support, feet shoulder width apart
2. Squat down while bending at the hips and knees only
3. Maintain consistent good posture; back straight, chest out, shoulders back
4. Lift slowly by straightening hips and knees, not your back. Keep consistence posture not twisting at the back.
5. Hold the load as close to the body as possible, this helps to maximise balance and strength
6. Take small steps, communication with whoever is helping you transport the object
7. When changing direction lead with hips, keeping shoulders in line with hips
8. When setting the object down, lower slowly squatting with the knees and hips only

Care and warning labels:

Care Instructions

- Dust the chair regularly with a soft cloth
- Keep the chair out of direct sunlight
- Use a coaster with any drinks paced on the elevated chair
- Keep water and other liquids from spilling on the chair. if spillage occurs, use a damp cloth to wipe the chair
- For any markings on steel, cover them up with a furniture marker pen

Warnings



- Do not lift the chair alone
- When lifting the chair use the correct form;

1. Keep a wide base of support, feet shoulder width apart
2. Squat down while bending at the hips and knees only
3. Maintain consistent good posture; back straight, chest out, shoulders back
4. Lift slowly by straightening hips and knees, not your back. Keep consistence posture not twisting at the back.
5. Hold the load as close to the body as possible, this helps to maximise balance and strength
6. Take small steps, communication with whoever is helping you transport the object
7. When changing direction lead with hips, keeping shoulders in line with hips
8. When setting the object down, lower slowly squatting with the knees and hips only

- Do not drag chair across the floor

Evaluation With Criteria

Criteria	How was this assessed with the product?	How well was this achieved (1-10)	My feedback	End user feedback
Constraints				
Is the final product long lasting and durable?	Over time evaluations were made to whether the product was suffering any marking or if signs of deterioration were present.	10/10	After assessing the various aspects of the chair over time for any signs of damage I concluded that the chair is exceptionally durable regarding both the finish and the strength of the chair. The durability of the chair allows it to complete its purpose to the best ability. It also means that the end users will be able to use this product for a long time which was an intended design feature.	Yes, the product appears to be very sturdy and after testing it myself I can attest to its durability.
Does the final product support both end users' weight?	This was assessed by having both end users test the product after initial tests for sufficient strength of the chair were done in the workshop. The end users made comments on the overall durability of the chair and any concerns were addressed.	10/10	After completing sufficient weight tests and assessing the strength of all the supports and various components of the chair, I can conclude that it supports both end user's weights with ease. No strain is put on the supports of the chair and the weight is evenly distributed. Having the centre of mass of the chair directly over the strongest part of the legs and by adding additions steel supports to the chair I am confident that the chair will continue to support both end users' weight.	Definitely, after testing the chair with my son all previous concerns were alleviated and I feel safe in the chair. The added supports, while not in the initial design, give me great confidence that the chair will hold mine and my sons weight especially if he bounces around.
Is the final product durable for repeated everyday use?	Following the completion of my product it was moved into the end users' home in order to test how the product handled everyday use and repeated usage.	9/10	The finishes applied to the chair and the care taken with the steel supports added additional strength to the product. This prevents the wood from being damaged and the extra supports add to the strength of the chair. The extra strength allows the chair to be used repeatedly without having to worry about damaging it.	Based on the finishes I am relatively assured that the product will withstand repeated usage in my home. The same goes for the durability of the chair, after testing it I can say that it appears very durable for everyday use. My only minor concern is that the painted finish on the legs may scuff and mark.
Does the product account for Logan's expected future growth?	This was assessed by checking all the steel supports and by taking measurements of the final product to check against Logan's future growth data.	7/10	While the durability of the product allows for additional weight expected with future growth, the only concern I have is with the products accessibility with Logan's expected	While the chair's design and construction allow for durability for any added weight, if Logan grows more then expected the comfort for both of us to use the chair may decrease.

			growth in the length of his legs. This could potentially create an awkward angle that Melissa needs to sit at due to the angle of the elevated chair. However the chair will still be perfectly functional, the ease of functionality is the only issue.	
Is the final product easily accessible?	This was checked though a physically assessment of the final product and testing with the end users their ease of getting out an into the chair repeatedly. They gave feedback on the accessibility.	7/10	After testing the chair, it was concluded that it was very accessible majority of the time. The only issue being that the main chair seat is only accessible from one side. The issue was raised that this could impact the choice in location for the chair. other than that, both end users are able to get in and out of the chair with ease.	Me and Logan both found it easy to get in and out of the chair from one side, however I noticed that the open side would need to be more accessible.
Is the final product multifunctional and can be used without Logan?	Again this was tested through physical assessments by having the end user complete a variety of tasks in the chair without Logan being in it, and feedback was gathered from observing and from the end user.	10/10	After evaluating this criteria, it was discovered that the chair is remarkably multifunctional and that the elevated chair creates a perfect height side table for the end user to use for a variety of tasks.	I was surprised by the multifunctional aspect of the chair, which I can use to hold my coffee or iPad when Logan isn't around.
Is the final product safe?	This was assessed with many methods due to this criterion being so crucial to the usability of the product. The first method was to converse with the end user and address any previous concerns they had about the safety of the product. The other method of evaluation was running weight and movement tests without the end user to properly ascertain a final conclusion on the safety of the product.	9/10	With safety being one of the major focuses for the end product both for the safety of the end users when using the chair but also for those moving around the chair. As mentioned in the previous criteria about the chair supporting both end users' weight, the chair is completely safe for the end users to use and the durability removes any concerns about end user safety. The only safety issue I see is with the pointed ends of the triangle legs. This can cause a hazard for people walking around the chair. This hazard causes the risk of tripping and accidentally kicking the legs. While this issue is minor it is still important to evaluate.	While I feel completely safe when sitting in the chair with Logan. The designer mentioned their concerns for the tripping hazard of the chair, I don't see this as anymore of an issue then with most other chairs in my house.
Considerations				

Product evaluation

While the chair is not perfect, it provides a well-functioning proof of concept for a chair that could be industrially manufactured as stated previously in my design profile. This product could help a lot of families and for that reason I believe that the final product is an absolute success. The empathetic design was formulated through a multitude of research which influenced the durability and accessibility of the chair.

Through initial conversations with the end user my design brief was formed which established the foundation for all decision to be made. Through the information gathered, background information of the end users, the conversations with the end user and the deign brief an evaluation criterion was created which would be used as a guide throughout the entire product design process. These processes ensured that the product was completed with all the end users wants and needs prioritised. This prompted the thorough research into all aspects of the product, including ergonomics and anthropometrics which helped to create a product that was suitable for the end users. After completing all research, initial visualisations were completed exploring a variety of shapes, forms and ideas in how to create the more suitable chair for the end users. Throughout the design option development and preferred design option stage a multitude of conversations were had with the end user to alter things to give them a design that best fitted their constraints and considerations. This initiated the use of new and emerging technology when creating a model in CAD. This process was beneficial to both me when designing and the end user as it provided a visual representation of the product. The model also assisted with creating the working drawings which later helped with the cut list and ordering of materials.

To begin planning a detailed production plan and Ghant chart were created to schedule all the steps into the allocated time. This planning kickstarted all other aspects of planning that needed to be done before construction could begin. Through the planning stages assessments were done for the level of risk in all planned production processes and with the quality measures. These helped to ensure that based on all the planning done, the product would be completed with the most minimal risk and that all precautions would be taken to achieve the highest quality product. Another key aspect of planning that took place before production commenced was material testing This included attempting smaller scale versions of production steps that were planned on taking place. The material and process testing provided a glimpse into some of the processes and how they could best be completed.

After all planning steps were completed, production began. Production kicked off smoothly with minimal modifications needing to be completed. Throughout the entirety of production, a multitude of modifications would need to be completed to not only fix any problems but also to improve upon the design in anyway possible. This included adding more supports, removing any unnecessary features of the product, and modifying processes to be more time efficient and reduce wastage. The production stage while often complex produced a final product that both the end user, and I are very happy with.

Once the product was finalised various stages of evaluations began. These included looking back upon the plan for the product and justifying the changes. This stage produced many lengthy conversations with the end user in relation to the evaluation criteria and the previously created criterion. While there are some aspects of the final product that both the end user and I have reservations about, the overall feedback is very positive and presents few areas for improvement. The main areas of improvement fit into three of the product design factors. These factors were crucial to the success for the final product and only had minor areas for improvement. The evaluation stage included care instructions and warnings for the end user and other people interacting with the product. These instructions and labels, conduct the users into the optimal ways to look after the product and any safety warnings that should look out for.

To conclude, through the various stages of developing this product a functional, durable and fit for purpose end product was created. The product not only fulfills the end users' needs but also achieves other vital aspects of importance to the end user. The main take away from this product is that there is potential for a product like this to benefit families with members with various disabilities on an industrial level. However, at this stage a product that helps one family have a slightly better life is a complete success.